Conditional Probability and Conditional Expectation

3

3.1 Introduction

One of the most useful concepts in probability theory is that of conditional probability and conditional expectation. The reason is twofold. First, in practice, we are often interested in calculating probabilities and expectations when some partial information is available; hence, the desired probabilities and expectations are conditional ones. Secondly, in calculating a desired probability or expectation it is often extremely useful to first "condition" on some appropriate random variable.

3.2 The Discrete Case

Recall that for any two events E and F, the conditional probability of E given F is defined, as long as $P(F) > 0$, by

$$P(E|F) = \frac{P(EF)}{P(F)}$$

Hence, if X and Y are discrete random variables, then it is natural to define the *conditional probability mass function* of X given that $Y = y$, by

$$\begin{aligned} p_{X|Y}(x|y) &= P\{X = x|Y = y\} \\ &= \frac{P\{X = x, Y = y\}}{P\{Y = y\}} \\ &= \frac{p(x, y)}{p_Y(y)} \end{aligned}$$

for all values of y such that $P\{Y = y\} > 0$. Similarly, the conditional probability distribution function of X given that $Y = y$ is defined, for all y such that $P\{Y = y\} > 0$, by

$$\begin{aligned} F_{X|Y}(x|y) &= P\{X \le x|Y = y\} \\ &= \sum_{a \le x} p_{X|Y}(a|y) \end{aligned}$$

Finally, the conditional expectation of X given that $Y = y$ is defined by

$$E[X|Y = y] = \sum_x x P\{X = x|Y = y\}$$

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$$= \sum_x x p_{X|Y}(x|y)$$

In other words, the definitions are exactly as before with the exception that everything is now conditional on the event that $Y = y$. If X is independent of Y, then the conditional mass function, distribution, and expectation are the same as the unconditional ones. This follows, since if X is independent of Y, then

$$\begin{aligned} p_{X|Y}(x|y) &= P\{X = x|Y = y\} \\ &= P\{X = x\} \end{aligned}$$

Example 3.1. Suppose that $p(x, y)$, the joint probability mass function of X and Y, is given by

$$p(1, 1) = 0.5, \quad p(1, 2) = 0.1, \quad p(2, 1) = 0.1, \quad p(2, 2) = 0.3$$

Calculate the conditional probability mass function of X given that $Y = 1$.

Solution: We first note that

$$p_Y(1) = \sum_x p(x, 1) = p(1, 1) + p(2, 1) = 0.6$$

Hence,

$$\begin{aligned} p_{X|Y}(1|1) &= P\{X = 1|Y = 1\} \\ &= \frac{P\{X = 1, Y = 1\}}{P\{Y = 1\}} \\ &= \frac{p(1, 1)}{p_Y(1)} \\ &= \frac{5}{6} \end{aligned}$$

Similarly,

$$p_{X|Y}(2|1) = \frac{p(2, 1)}{p_Y(1)} = \frac{1}{6}$$ ■

Example 3.2. If X_1 and X_2 are independent binomial random variables with respective parameters (n_1, p) and (n_2, p), calculate the conditional probability mass function of X_1 given that $X_1 + X_2 = m$.

Solution: With $q = 1 - p$,

$$\begin{aligned} P\{X_1 = k|X_1 + X_2 = m\} &= \frac{P\{X_1 = k, X_1 + X_2 = m\}}{P\{X_1 + X_2 = m\}} \\ &= \frac{P\{X_1 = k, X_2 = m - k\}}{P\{X_1 + X_2 = m\}} \end{aligned}$$

$$= \frac{P\{X_1 = k\}P\{X_2 = m-k\}}{P\{X_1 + X_2 = m\}}$$
$$= \frac{\binom{n_1}{k} p^k q^{n_1-k} \binom{n_2}{m-k} p^{m-k} q^{n_2-m+k}}{\binom{n_1+n_2}{m} p^m q^{n_1+n_2-m}}$$

where we have used that $X_1 + X_2$ is a binomial random variable with parameters $(n_1 + n_2, p)$ (see Example 2.44). Thus, the conditional probability mass function of X_1, given that $X_1 + X_2 = m$, is

$$P\{X_1 = k | X_1 + X_2 = m\} = \frac{\binom{n_1}{k}\binom{n_2}{m-k}}{\binom{n_1+n_2}{m}} \tag{3.1}$$

The distribution given by Eq. (3.1), first seen in Example 2.35, is known as the *hypergeometric* distribution. It is the distribution of the number of blue balls that are chosen when a sample of m balls is randomly chosen from an urn that contains n_1 blue and n_2 red balls. (To intuitively see why the conditional distribution is hypergeometric, consider $n_1 + n_2$ independent trials that each result in a success with probability p; let X_1 represent the number of successes in the first n_1 trials and let X_2 represent the number of successes in the final n_2 trials. Because all trials have the same probability of being a success, each of the $\binom{n_1+n_2}{m}$ subsets of m trials is equally likely to be the success trials; thus, the number of the m success trials that are among the first n_1 trials is a hypergeometric random variable.) ■

Example 3.3. If X and Y are independent Poisson random variables with respective means λ_1 and λ_2, calculate the conditional expected value of X given that $X + Y = n$.

Solution: Let us first calculate the conditional probability mass function of X given that $X + Y = n$. We obtain

$$P\{X = k | X + Y = n\} = \frac{P\{X = k, X + Y = n\}}{P\{X + Y = n\}}$$
$$= \frac{P\{X = k, Y = n - k\}}{P\{X + Y = n\}}$$
$$= \frac{P\{X = k\}P\{Y = n - k\}}{P\{X + Y = n\}}$$

where the last equality follows from the assumed independence of X and Y. Recalling (see Example 2.37) that $X + Y$ has a Poisson distribution with mean $\lambda_1 + \lambda_2$, the preceding equation equals

$$P\{X = k | X + Y = n\} = \frac{e^{-\lambda_1}\lambda_1^k}{k!} \frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!} \left[\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}\right]^{-1}$$

$$= \frac{n!}{(n-k)!k!} \frac{\lambda_1^k \lambda_2^{n-k}}{(\lambda_1+\lambda_2)^n}$$
$$= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}$$

In other words, the conditional distribution of X given that $X+Y=n$ is the binomial distribution with parameters n and $\lambda_1/(\lambda_1+\lambda_2)$. Hence,

$$E\{X|X+Y=n\} = n\frac{\lambda_1}{\lambda_1+\lambda_2}$$ ■

Conditional expectations possess all of the properties of ordinary expectations. For example such identities such as

$$E\left[\sum_{i=1}^{n} X_i | Y=y\right] = \sum_{i=1}^{n} E[X_i|Y=y]$$
$$E[h(X)|Y=y] = \sum_x h(x)P(X=x|Y=y)$$

remain valid.

Example 3.4. There are n components. On a rainy day, component i will function with probability p_i; on a nonrainy day, component i will function with probability q_i, for $i=1,\ldots,n$. It will rain tomorrow with probability α. Calculate the conditional expected number of components that function tomorrow, given that it rains.

Solution: Let

$$X_i = \begin{cases} 1, & \text{if component } i \text{ functions tomorrow} \\ 0, & \text{otherwise} \end{cases}$$

Then, with Y defined to equal 1 if it rains tomorrow, and 0 otherwise, the desired conditional expectation is obtained as follows.

$$E\left[\sum_{i=1}^{n} X_i | Y=1\right] = \sum_{i=1}^{n} E[X_i|Y=1]$$
$$= \sum_{i=1}^{n} p_i$$ ■

3.3 The Continuous Case

If X and Y have a joint probability density function $f(x,y)$, then the *conditional probability density function* of X, given that $Y=y$, is defined for all values of y such

that $f_Y(y) > 0$, by

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

To motivate this definition, multiply the left side by dx and the right side by $(dx\, dy)/dy$ to get

$$\begin{aligned} f_{X|Y}(x|y)\, dx &= \frac{f(x, y)\, dx\, dy}{f_Y(y)\, dy} \\ &\approx \frac{P\{x \le X \le x + dx, y \le Y \le y + dy\}}{P\{y \le Y \le y + dy\}} \\ &= P\{x \le X \le x + dx | y \le Y \le y + dy\} \end{aligned}$$

In other words, for small values dx and dy, $f_{X|Y}(x|y)\, dx$ is approximately the conditional probability that X is between x and $x + dx$ given that Y is between y and $y + dy$.

The *conditional expectation* of X, given that $Y = y$, is defined for all values of y such that $f_Y(y) > 0$, by

$$E[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx$$

Example 3.5. Suppose the joint density of X and Y is given by

$$f(x, y) = \begin{cases} 6xy(2 - x - y), & 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Compute the conditional expectation of X given that $Y = y$, where $0 < y < 1$.

Solution: We first compute the conditional density

$$\begin{aligned} f_{X|Y}(x|y) &= \frac{f(x, y)}{f_Y(y)} \\ &= \frac{6xy(2 - x - y)}{\int_0^1 6xy(2 - x - y)\, dx} \\ &= \frac{6xy(2 - x - y)}{y(4 - 3y)} \\ &= \frac{6x(2 - x - y)}{4 - 3y} \end{aligned}$$

Hence,

$$\begin{aligned} E[X|Y = y] &= \int_0^1 \frac{6x^2(2 - x - y)\, dx}{4 - 3y} \\ &= \frac{(2 - y)2 - \frac{6}{4}}{4 - 3y} \end{aligned}$$

$$= \frac{5-4y}{8-6y} \qquad \blacksquare$$

Example 3.6 (The t-Distribution)**.** If Z and Y are independent, with Z having a standard normal distribution and Y having a chi-squared distribution with n degrees of freedom, then the random variable T defined by

$$T = \frac{Z}{\sqrt{Y/n}} = \sqrt{n}\frac{Z}{\sqrt{Y}}$$

is said to be a *t-random variable* with n degrees of freedom. To compute its density function, we first derive the conditional density of T given that $Y = y$. Because Z and Y are independent, the conditional distribution of T given that $Y = y$ is the distribution of $\sqrt{n/y}Z$, which is normal with mean 0 and variance n/y. Hence, the conditional density function of T given that $Y = y$ is

$$f_{T|Y}(t|y) = \frac{1}{\sqrt{2\pi n/y}} e^{-t^2 y/2n} = \frac{y^{1/2}}{\sqrt{2\pi n}} e^{-t^2 y/2n}, \quad -\infty < t < \infty$$

Using the preceding, along with the following formula for the chi-squared density derived in Exercise 87 of Chapter 2,

$$f_Y(y) = \frac{e^{-y/2} y^{n/2-1}}{2^{n/2}\Gamma(n/2)}, \quad y > 0$$

we obtain the density function of T:

$$f_T(t) = \int_0^\infty f_{T,Y}(t, y)\, dy = \int_0^\infty f_{T|Y}(t|y) f_Y(y)\, dy$$

With

$$K = \frac{1}{\sqrt{\pi n}\, 2^{(n+1)/2}\Gamma(n/2)}, \quad c = \frac{t^2+n}{2n} = \frac{1}{2}\left(1 + \frac{t^2}{n}\right)$$

this yields

$$\begin{aligned} f_T(t) &= \frac{1}{K}\int_0^\infty e^{-cy} y^{(n-1)/2}\, dy \\ &= \frac{c^{-(n+1)/2}}{K}\int_0^\infty e^{-x} x^{(n-1)/2}\, dx \quad \text{(by letting } x = cy\text{)} \\ &= \frac{c^{-(n+1)/2}}{K}\Gamma\left(\frac{n+1}{2}\right) \\ &= \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n}\,\Gamma(\frac{n}{2})}\left(1 + \frac{t^2}{n}\right)^{-(n+1)/2}, \quad -\infty < t < \infty \end{aligned} \qquad \blacksquare$$

Example 3.7. The joint density of X and Y is given by

$$f(x,y) = \begin{cases} \frac{1}{2}ye^{-xy}, & 0 < x < \infty,\ 0 < y < 2 \\ 0, & \text{otherwise} \end{cases}$$

What is $E[e^{X/2}|Y=1]$?

Solution: The conditional density of X, given that $Y=1$, is given by

$$\begin{aligned} f_{X|Y}(x|1) &= \frac{f(x,1)}{f_Y(1)} \\ &= \frac{\frac{1}{2}e^{-x}}{\int_0^\infty \frac{1}{2}e^{-x}\,dx} = e^{-x} \end{aligned}$$

Hence, by Proposition 2.1,

$$\begin{aligned} E\left[e^{X/2}|Y=1\right] &= \int_0^\infty e^{x/2} f_{X|Y}(x|1)\,dx \\ &= \int_0^\infty e^{x/2}e^{-x}\,dx \\ &= 2 \end{aligned}$$

■

Example 3.8. Let X_1 and X_2 be independent exponential random variables with rates μ_1 and μ_2. Find the conditional density of X_1 given that $X_1 + X_2 = t$.

Solution: To begin, let us first note that if $f(x,y)$ is the joint density of X, Y, then the joint density of X and $X+Y$ is

$$f_{X,X+Y}(x,t) = f(x,t-x)$$

which is easily seen by noting that the Jacobian of the transformation

$$g_1(x,y) = x, \quad g_2(x,y) = x + y$$

is equal to 1.
Applying the preceding to our example yields

$$\begin{aligned} f_{X_1|X_1+X_2}(x|t) &= \frac{f_{X_1,X_1+X_2}(x,t)}{f_{X_1+X_2}(t)} \\ &= \frac{\mu_1 e^{-\mu_1 x}\mu_2 e^{-\mu_2(t-x)}}{f_{X_1+X_2}(t)}, \quad 0 \le x \le t \\ &= Ce^{-(\mu_1-\mu_2)x}, \quad 0 \le x \le t \end{aligned}$$

where

$$C = \frac{\mu_1\mu_2 e^{-\mu_2 t}}{f_{X_1+X_2}(t)}$$

Now, if $\mu_1 = \mu_2$, then

$$f_{X_1|X_1+X_2}(x|t) = C, \quad 0 \le x \le t$$

yielding that $C = 1/t$ and that X_1 given $X_1 + X_2 = t$ is uniformly distributed on $(0, t)$. On the other hand, if $\mu_1 \neq \mu_2$, then we use

$$1 = \int_0^t f_{X_1|X_1+X_2}(x|t)dx = \frac{C}{\mu_1 - \mu_2}\left(1 - e^{-(\mu_1-\mu_2)t}\right)$$

to obtain

$$C = \frac{\mu_1 - \mu_2}{1 - e^{-(\mu_1-\mu_2)t}}$$

thus yielding the result:

$$f_{X_1|X_1+X_2}(x|t) = \frac{(\mu_1 - \mu_2)e^{-(\mu_1-\mu_2)x}}{1 - e^{-(\mu_1-\mu_2)t}}$$

An interesting byproduct of our analysis is that

$$f_{X_1+X_2}(t) = \frac{\mu_1\mu_2 e^{-\mu_2 t}}{C} = \begin{cases} \mu^2 t e^{-\mu t}, & \text{if } \mu_1 = \mu_2 = \mu \\ \frac{\mu_1\mu_2(e^{-\mu_2 t} - e^{-\mu_1 t})}{\mu_1 - \mu_2}, & \text{if } \mu_1 \neq \mu_2 \end{cases}$$ ■

3.4 Computing Expectations by Conditioning

Let us denote by $E[X|Y]$ that function of the random variable Y whose value at $Y = y$ is $E[X|Y = y]$. Note that $E[X|Y]$ is itself a random variable. An extremely important property of conditional expectation is that for all random variables X and Y

$$E[X] = E\big[E[X|Y]\big] \tag{3.2}$$

If Y is a discrete random variable, then Eq. (3.2) states that

$$E[X] = \sum_y E[X|Y = y]P\{Y = y\} \tag{3.2a}$$

while if Y is continuous with density $f_Y(y)$, then Eq. (3.2) says that

$$E[X] = \int_{-\infty}^{\infty} E[X|Y = y]f_Y(y)\,dy \tag{3.2b}$$

We now give a proof of Eq. (3.2) in the case where X and Y are both discrete random variables.

Proof of Eq. (3.2) When X and Y Are Discrete. We must show that

$$E[X] = \sum_y E[X|Y=y]P\{Y=y\} \tag{3.3}$$

Now, the right side of the preceding can be written

$$\begin{aligned}
\sum_y E[X|Y=y]P\{Y=y\} &= \sum_y \sum_x x P\{X=x|Y=y\}P\{Y=y\} \\
&= \sum_y \sum_x x \frac{P\{X=x, Y=y\}}{P\{Y=y\}} P\{Y=y\} \\
&= \sum_y \sum_x x P\{X=x, Y=y\} \\
&= \sum_x x \sum_y P\{X=x, Y=y\} \\
&= \sum_x x P\{X=x\} \\
&= E[X]
\end{aligned}$$

and the result is obtained. ■

One way to understand Eq. (3.3) is to interpret it as follows. It states that to calculate $E[X]$ we may take a weighted average of the conditional expected value of X given that $Y = y$, each of the terms $E[X|Y=y]$ being weighted by the probability of the event on which it is conditioned.

The following examples will indicate the usefulness of Eq. (3.2).

Example 3.9. Sam will read either one chapter of his probability book or one chapter of his history book. If the number of misprints in a chapter of his probability book is Poisson distributed with mean 2 and if the number of misprints in his history chapter is Poisson distributed with mean 5, then assuming Sam is equally likely to choose either book, what is the expected number of misprints that Sam will come across?

Solution: Let X be the number of misprints. Because it would be easy to compute $E[X]$ if we know which book Sam chooses, let

$$Y = \begin{cases} 1, & \text{if Sam chooses his history book} \\ 2, & \text{if chooses his probability book} \end{cases}$$

Conditioning on Y yields

$$\begin{aligned}
E[X] &= E[X|Y=1]P\{Y=1\} + E[X|Y=2]P\{Y=2\} \\
&= 5\left(\tfrac{1}{2}\right) + 2\left(\tfrac{1}{2}\right) \\
&= \tfrac{7}{2}
\end{aligned}$$

■

Example 3.10 (The Expectation of the Sum of a Random Number of Random Variables)**.** Suppose that the expected number of accidents per week at an industrial plant is four. Suppose also that the numbers of workers injured in each accident are independent random variables with a common mean of 2. Assume also that the number of workers injured in each accident is independent of the number of accidents that occur. What is the expected number of injuries during a week?

Solution: Letting N denote the number of accidents and X_i the number injured in the ith accident, $i = 1, 2, \ldots$, then the total number of injuries can be expressed as $\sum_{i=1}^{N} X_i$. Hence, we need to compute the expected value of the sum of a random number of random variables. Because it is easy to compute the expected value of the sum of a fixed number of random variables, let us try conditioning on N. This yields

$$E\left[\sum_{1}^{N} X_i\right] = E\left[E\left[\sum_{1}^{N} X_i | N\right]\right]$$

But

$$\begin{aligned} E\left[\sum_{1}^{N} X_i | N = n\right] &= E\left[\sum_{1}^{n} X_i | N = n\right] \\ &= E\left[\sum_{1}^{n} X_i\right] \quad \text{by the independence of } X_i \text{ and } N \\ &= nE[X] \end{aligned}$$

which yields

$$E\left[\sum_{i=1}^{N} X_i | N\right] = NE[X]$$

and thus

$$E\left[\sum_{i=1}^{N} X_i\right] = E\big[NE[X]\big] = E[N]E[X]$$

Therefore, in our example, the expected number of injuries during a week equals $4 \times 2 = 8$. ■

The random variable $\sum_{i=1}^{N} X_i$, equal to the sum of a random number N of independent and identically distributed random variables that are also independent of N, is called a *compound random variable*. As just shown in Example 3.10, the expected value of a compound random variable is $E[X]E[N]$. Its variance will be derived in Example 3.20.

If there is some random variable Y such that it would be easy to compute $E[X]$ if we knew the value of Y, then conditioning on Y is likely to be a good strategy for determining $E[X]$. When there is no obvious random variable to condition on, it often turns out to be useful to condition on the first thing that occurs. This is illustrated in the following two examples.

Example 3.11 (The Mean of a Geometric Distribution)**.** A coin, having probability p of coming up heads, is to be successively flipped until the first head appears. What is the expected number of flips required?

Solution: Let N be the number of flips required, and let

$$Y = \begin{cases} 1, & \text{if the first flip results in a head} \\ 0, & \text{if the first flip results in a tail} \end{cases}$$

Now,

$$\begin{aligned} E[N] &= E[N|Y=1]P\{Y=1\} + E[N|Y=0]P\{Y=0\} \\ &= pE[N|Y=1] + (1-p)E[N|Y=0] \end{aligned} \tag{3.4}$$

However,

$$E[N|Y=1] = 1, \quad E[N|Y=0] = 1 + E[N] \tag{3.5}$$

To see why Eq. (3.5) is true, consider $E[N|Y=1]$. Since $Y=1$, we know that the first flip resulted in heads and so, clearly, the expected number of flips required is 1. On the other hand if $Y=0$, then the first flip resulted in tails. However, since the successive flips are assumed independent, it follows that, after the first tail, the expected additional number of flips until the first head is just $E[N]$. Hence $E[N|Y=0] = 1 + E[N]$. Substituting Eq. (3.5) into Eq. (3.4) yields

$$E[N] = p + (1-p)(1 + E[N])$$

or

$$E[N] = 1/p$$ ■

Because the random variable N is a geometric random variable with probability mass function $p(n) = p(1-p)^{n-1}$, its expectation could easily have been computed from $E[N] = \sum_1^\infty np(n)$ without recourse to conditional expectation. However, if you attempt to obtain the solution to our next example without using conditional expectation, you will quickly learn what a useful technique "conditioning" can be.

Example 3.12. A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to his mine after five hours. Assuming that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until the miner reaches safety?

Solution: Let X denote the time until the miner reaches safety, and let Y denote the door he initially chooses. Now,

$$\begin{aligned} E[X] &= E[X|Y=1]P\{Y=1\} + E[X|Y=2]P\{Y=2\} \\ &\quad + E[X|Y=3]P\{Y=3\} \\ &= \tfrac{1}{3}\big(E[X|Y=1] + E[X|Y=2] + E[X|Y=3]\big) \end{aligned}$$

However,

$$\begin{aligned} &E[X|Y=1] = 2, \\ &E[X|Y=2] = 3 + E[X], \\ &E[X|Y=3] = 5 + E[X], \end{aligned} \tag{3.6}$$

To understand why this is correct consider, for instance, $E[X|Y=2]$, and reason as follows. If the miner chooses the second door, then he spends three hours in the tunnel and then returns to the mine. But once he returns to the mine the problem is as before, and hence his expected additional time until safety is just $E[X]$. Hence $E[X|Y=2] = 3 + E[X]$. The argument behind the other equalities in Eq. (3.6) is similar. Hence,

$$E[X] = \tfrac{1}{3}(2 + 3 + E[X] + 5 + E[X]) \quad \text{or} \quad E[X] = 10 \qquad \blacksquare$$

Example 3.13 (Multinomial Covariances)**.** Consider n independent trials, each of which results in one of the outcomes $1, \ldots, r$, with respective probabilities $p_1, \ldots, p_r$, $\sum_{i=1}^{r} p_i = 1$. If we let N_i denote the number of trials that result in outcome i, then $(N_1, \ldots, N_r)$ is said to have a *multinomial distribution*. For $i \neq j$, let us compute

$$\mathrm{Cov}(N_i, N_j) = E[N_i N_j] - E[N_i]E[N_j]$$

Because each trial independently results in outcome i with probability p_i, it follows that N_i is binomial with parameters (n, p_i), giving that $E[N_i]E[N_j] = n^2 p_i p_j$. To compute $E[N_i N_j]$, condition on N_i to obtain

$$\begin{aligned} E[N_i N_j] &= \sum_{k=0}^{n} E[N_i N_j | N_i = k] P(N_i = k) \\ &= \sum_{k=0}^{n} k E[N_j | N_i = k] P(N_i = k) \end{aligned}$$

Now, given that k of the n trials result in outcome i, each of the other $n - k$ trials independently results in outcome j with probability

$$P(j|\text{not } i) = \frac{p_j}{1 - p_i}$$

thus showing that the conditional distribution of N_j, given that $N_i = k$, is binomial with parameters $(n-k, \frac{p_j}{1-p_i})$. Using this yields

$$\begin{aligned} E[N_i N_j] &= \sum_{k=0}^{n} k(n-k)\frac{p_j}{1-p_i} P(N_i = k) \\ &= \frac{p_j}{1-p_i}\left(n\sum_{k=0}^{n} kP(N_i = k) - \sum_{k=0}^{n} k^2 P(N_i = k)\right) \\ &= \frac{p_j}{1-p_i}(nE[N_i] - E[N_i^2]) \end{aligned}$$

Because N_i is binomial with parameters (n, p_i)

$$E[N_i^2] = \mathrm{Var}(N_i) + (E[N_i])^2 = np_i(1-p_i) + (np_i)^2$$

Hence,

$$\begin{aligned} E[N_i N_j] &= \frac{p_j}{1-p_i}[n^2 p_i - np_i(1-p_i) - n^2 p_i^2] \\ &= \frac{np_i p_j}{1-p_i}[n - np_i - (1-p_i)] \\ &= n(n-1)p_i p_j \end{aligned}$$

which yields the result

$$\mathrm{Cov}(N_i, N_j) = n(n-1)p_i p_j - n^2 p_i p_j = -np_i p_j$$ ■

Example 3.14 (The Matching Rounds Problem)**.** Suppose in Example 2.30 that those choosing their own hats depart, while the others (those without a match) put their selected hats in the center of the room, mix them up, and then reselect. Also, suppose that this process continues until each individual has his own hat.

(a) Find $E[R_n]$ where R_n is the number of rounds that are necessary when n individuals are initially present.

(b) Find $E[S_n]$ where S_n is the total number of selections made by the n individuals, $n \geq 2$.

(c) Find the expected number of false selections made by one of the n people, $n \geq 2$.

Solution: (a) It follows from the results of Example 2.30 that no matter how many people remain there will, on average, be one match per round. Hence, one might suggest that $E[R_n] = n$. This turns out to be true, and an induction proof will now be given. Because it is obvious that $E[R_1] = 1$, assume that $E[R_k] = k$ for $k = 1, \ldots, n-1$. To compute $E[R_n]$, start by conditioning on X_n, the number of matches that occur in the first round. This gives

$$E[R_n] = \sum_{i=0}^{n} E[R_n | X_n = i]P\{X_n = i\}$$

Now, given a total of i matches in the initial round, the number of rounds needed will equal 1 plus the number of rounds that are required when $n - i$ persons are to be matched with their hats. Therefore,

$$\begin{aligned}
E[R_n] &= \sum_{i=0}^{n} (1 + E[R_{n-i}]) P\{X_n = i\} \\
&= 1 + E[R_n] P\{X_n = 0\} + \sum_{i=1}^{n} E[R_{n-i}] P\{X_n = i\} \\
&= 1 + E[R_n] P\{X_n = 0\} + \sum_{i=1}^{n} (n - i) P\{X_n = i\} \\
&\qquad \text{by the induction hypothesis} \\
&= 1 + E[R_n] P\{X_n = 0\} + n(1 - P\{X_n = 0\}) - E[X_n] \\
&= E[R_n] P\{X_n = 0\} + n(1 - P\{X_n = 0\})
\end{aligned}$$

where the final equality used the result, established in Example 2.30, that $E[X_n] = 1$. Since the preceding equation implies that $E[R_n] = n$, the result is proven.

(b) For $n \geq 2$, conditioning on X_n, the number of matches in round 1, gives

$$\begin{aligned}
E[S_n] &= \sum_{i=0}^{n} E[S_n | X_n = i] P\{X_n = i\} \\
&= \sum_{i=0}^{n} (n + E[S_{n-i}]) P\{X_n = i\} \\
&= n + \sum_{i=0}^{n} E[S_{n-i}] P\{X_n = i\}
\end{aligned}$$

where $E[S_0] = 0$. To solve the preceding equation, rewrite it as

$$E[S_n] = n + E[S_{n-X_n}]$$

Now, if there were *exactly* one match in each round, then it would take a total of $1 + 2 + \cdots + n = n(n + 1)/2$ selections. Thus, let us try a solution of the form $E[S_n] = an + bn^2$. For the preceding equation to be satisfied by a solution of this type, for $n \geq 2$, we need

$$an + bn^2 = n + E\left[a(n - X_n) + b(n - X_n)^2\right]$$

or, equivalently,

$$an + bn^2 = n + a(n - E[X_n]) + b\left(n^2 - 2nE[X_n] + E\left[X_n^2\right]\right)$$

Now, using the results of Example 2.30 and Exercise 72 of Chapter 2 that $E[X_n] = \text{Var}(X_n) = 1$, the preceding will be satisfied if

$$an + bn^2 = n + an - a + bn^2 - 2nb + 2b$$

and this will be valid provided that $b = 1/2, a = 1$. That is,

$$E[S_n] = n + n^2/2$$

satisfies the recursive equation for $E[S_n]$.

The formal proof that $E[S_n] = n + n^2/2, n \geq 2$, is obtained by induction on n. It is true when $n = 2$ (since, in this case, the number of selections is twice the number of rounds and the number of rounds is a geometric random variable with parameter $p = 1/2$). Now, the recursion gives

$$E[S_n] = n + E[S_n]P\{X_n = 0\} + \sum_{i=1}^{n} E[S_{n-i}]P\{X_n = i\}$$

Hence, upon assuming that $E[S_0] = E[S_1] = 0$, $E[S_k] = k + k^2/2$, for $k = 2, \ldots, n-1$ and using that $P\{X_n = n-1\} = 0$, we see that

$$\begin{aligned} E[S_n] &= n + E[S_n]P\{X_n = 0\} + \sum_{i=1}^{n}[n - i + (n-i)^2/2]P\{X_n = i\} \\ &= n + E[S_n]P\{X_n = 0\} + (n + n^2/2)(1 - P\{X_n = 0\}) \\ &\quad - (n+1)E[X_n] + E[X_n^2]/2 \end{aligned}$$

Substituting the identities $E[X_n] = 1$, $E[X_n^2] = 2$ in the preceding shows that

$$E[S_n] = n + n^2/2$$

and the induction proof is complete.
(c) If we let C_j denote the number of hats chosen by person j, $j = 1, \ldots, n$ then

$$\sum_{j=1}^{n} C_j = S_n$$

Taking expectations, and using the fact that each C_j has the same mean, yields the result

$$E[C_j] = E[S_n]/n = 1 + n/2$$

Hence, the expected number of false selections by person j is

$$E[C_j - 1] = n/2.$$

■

Example 3.15. Independent trials, each of which is a success with probability p, are performed until there are k consecutive successes. What is the mean number of necessary trials?

Solution: Let N_k denote the number of necessary trials to obtain k consecutive successes, and let $M_k = E[N_k]$. We will determine M_k by deriving and then solving a recursive equation that it satisfies. To begin, write

$$N_k = N_{k-1} + A_{k-1,k}$$

where N_{k-1} is the number of trials needed for $k-1$ consecutive successes, and $A_{k-1,k}$ is the number of additional trials needed to go from having $k-1$ successes in a row to having k in a row. Taking expectations gives that,

$$M_k = M_{k-1} + E[A_{k-1,k}]$$

To determine $E[A_{k-1,k}]$, condition on the next trial after there have been $k-1$ successes in a row. If it is a success then that gives k in a row and no additional trials after that are needed; if it is a failure then at that point we are starting all over and so the expected additional number from then on would be $E[N_k]$. Thus,

$$E[A_{k-1,k}] = 1 \cdot p + (1 + M_k)(1 - p) = 1 + (1 - p)M_k$$

giving that

$$M_k = M_{k-1} + 1 + (1 - p)M_k$$

or

$$M_k = \frac{1}{p} + \frac{M_{k-1}}{p}$$

Since N_1, the time of the first success, is geometric with parameter p, we see that

$$M_1 = \frac{1}{p}$$

and, recursively

$$M_2 = \frac{1}{p} + \frac{1}{p^2},$$
$$M_3 = \frac{1}{p} + \frac{1}{p^2} + \frac{1}{p^3}$$

and, in general,

$$M_k = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k}$$

■

Example 3.16. Consider a gambler who in each game is equally likely to either win or lose 1, independent of the results from earlier games. Starting with a fortune of i, find m_i, the mean number of games until the gambler's fortune is either 0 or n, where $0 \le i \le n$.

Solution: Let N denote the number of games until the gambler's fortune is either 0 or n, and let S_i denote the event that the gambler starts with a fortune of i. To obtain an expression for $m_i = E[N|S_i]$, condition on the result of the first game. With W being the event that the first game is a win, and L that it is a loss, this yields that for $i = 1, \ldots, n-1$

$$\begin{aligned} m_i &= E[N|S_i] \\ &= E[N|S_iW]\,P(W|S_i) + E[N|S_iL]\,P(L|S_i) \\ &= (1+m_{i+1})\frac{1}{2} + (1+m_{i-1})\frac{1}{2} \\ &= 1 + \frac{1}{2}m_{i-1} + \frac{1}{2}m_{i+1}, \quad i = 1, \ldots, n-1 \end{aligned}$$

Using that $m_0 = 0$, the preceding can be rewritten as

$$\begin{aligned} m_2 &= 2(m_1 - 1) \\ m_{i+1} &= 2(m_i - 1) - m_{i-1}, \quad i = 2, \ldots, n-1 \end{aligned}$$

Letting $i = 2$ in the preceding yields that

$$\begin{aligned} m_3 &= 2m_2 - 2 - m_1 \\ &= 4m_1 - 4 - 2 - m_1 \\ &= 3(m_1 - 2) \end{aligned}$$

A check of m_4 shows a similar result, and indeed it is easily shown by induction that

$$m_i = i(m_1 - i + 1), \quad i = 2, \ldots, n$$

Using that $m_n = 0$, the preceding yields that $0 = n(m_1 - n + 1)$. Thus, $m_1 = n - 1$, and

$$m_i = i(n-i), \qquad i = 1, \ldots, n-1$$ ■

Example 3.17 (Analyzing the Quick-Sort Algorithm)**.** Suppose we are given a set of n distinct values—$x_1, \ldots, x_n$—and we desire to put these values in increasing order or, as it is commonly called, to *sort* them. An efficient procedure for accomplishing this is the quick-sort algorithm, which is defined recursively as follows: When $n = 2$ the algorithm compares the two values and puts them in the appropriate order. When $n > 2$ it starts by choosing at random one of the n values—say, x_i—and then compares each of the other $n - 1$ values with x_i, noting which are smaller and which are larger

than x_i. Letting S_i denote the set of elements smaller than x_i, and $\bar{S}_i$ the set of elements greater than x_i, the algorithm now sorts the set S_i and the set $\bar{S}_i$. The final ordering, therefore, consists of the ordered set of the elements in S_i, then x_i, and then the ordered set of the elements in $\bar{S}_i$. For instance, suppose that the set of elements is 10, 5, 8, 2, 1, 4, 7. We start by choosing one of these values at random (that is, each of the 7 values has probability of $\frac{1}{7}$ of being chosen). Suppose, for instance, that the value 4 is chosen. We then compare 4 with each of the other six values to obtain

$$\{2, 1\}, 4, \{10, 5, 8, 7\}$$

We now sort the set $\{2, 1\}$ to obtain

$$1, 2, 4, \{10, 5, 8, 7\}$$

Next we choose a value at random from $\{10, 5, 8, 7\}$—say 7 is chosen—and compare each of the other three values with 7 to obtain

$$1, 2, 4, 5, 7, \{10, 8\}$$

Finally, we sort $\{10, 8\}$ to end up with

$$1, 2, 4, 5, 7, 8, 10$$

One measure of the effectiveness of this algorithm is the expected number of comparisons that it makes. Let us denote by M_n the expected number of comparisons needed by the quick-sort algorithm to sort a set of n distinct values. To obtain a recursion for M_n we condition on the rank of the initial value selected to obtain

$$M_n = \sum_{j=1}^{n} E[\text{number of comparisons}|\text{value selected is } j\text{th smallest}]\frac{1}{n}$$

Now, if the initial value selected is the jth smallest, then the set of values smaller than it is of size $j-1$, and the set of values greater than it is of size $n-j$. Hence, as $n-1$ comparisons with the initial value chosen must be made, we see that

$$\begin{aligned} M_n &= \sum_{j=1}^{n}(n-1+M_{j-1}+M_{n-j})\frac{1}{n} \\ &= n-1+\frac{2}{n}\sum_{k=1}^{n-1}M_k \quad (\text{since } M_0=0) \end{aligned}$$

or, equivalently,

$$nM_n = n(n-1)+2\sum_{k=1}^{n-1}M_k$$

To solve the preceding, note that upon replacing n by $n+1$ we obtain

$$(n+1)M_{n+1} = (n+1)n + 2\sum_{k=1}^{n} M_k$$

Hence, upon subtraction,

$$(n+1)M_{n+1} - nM_n = 2n + 2M_n$$

or

$$(n+1)M_{n+1} = (n+2)M_n + 2n$$

Therefore,

$$\frac{M_{n+1}}{n+2} = \frac{2n}{(n+1)(n+2)} + \frac{M_n}{n+1}$$

Iterating this gives

$$\begin{aligned}\frac{M_{n+1}}{n+2} &= \frac{2n}{(n+1)(n+2)} + \frac{2(n-1)}{n(n+1)} + \frac{M_{n-1}}{n} \\ &= \cdots \\ &= 2\sum_{k=0}^{n-1} \frac{n-k}{(n+1-k)(n+2-k)} \quad \text{since } M_1 = 0\end{aligned}$$

Hence,

$$\begin{aligned}M_{n+1} &= 2(n+2)\sum_{k=0}^{n-1} \frac{n-k}{(n+1-k)(n+2-k)} \\ &= 2(n+2)\sum_{i=1}^{n} \frac{i}{(i+1)(i+2)}, \quad n \geq 1\end{aligned}$$

Using the identity $i/(i+1)(i+2) = 2/(i+2) - 1/(i+1)$, we can approximate M_{n+1} for large n as follows:

$$\begin{aligned}M_{n+1} &= 2(n+2)\left[\sum_{i=1}^{n} \frac{2}{i+2} - \sum_{i=1}^{n} \frac{1}{i+1}\right] \\ &\sim 2(n+2)\left[\int_3^{n+2} \frac{2}{x}\,dx - \int_2^{n+1} \frac{1}{x}\,dx\right] \\ &= 2(n+2)[2\log(n+2) - \log(n+1) + \log 2 - 2\log 3] \\ &= 2(n+2)\left[\log(n+2) + \log\frac{n+2}{n+1} + \log 2 - 2\log 3\right] \\ &\sim 2(n+2)\log(n+2)\end{aligned}$$

■

Although we usually employ the conditional expectation identity to more easily enable us to compute an unconditional expectation, in our next example we show how it can sometimes be used to obtain the conditional expectation.

Example 3.18. In the match problem of Example 2.30 involving $n, n > 1$, individuals, find the conditional expected number of matches given that the first person did not have a match.

Solution: Let X denote the number of matches, and let X_1 equal 1 if the first person has a match and 0 otherwise. Then,

$$\begin{aligned} E[X] &= E[X|X_1 = 0]P\{X_1 = 0\} + E[X|X_1 = 1]P\{X_1 = 1\} \\ &= E[X|X_1 = 0]\frac{n-1}{n} + E[X|X_1 = 1]\frac{1}{n} \end{aligned}$$

But, from Example 2.30

$$E[X] = 1$$

Moreover, given that the first person has a match, the expected number of matches is equal to 1 plus the expected number of matches when $n - 1$ people select among their own $n - 1$ hats, showing that

$$E[X|X_1 = 1] = 2$$

Therefore, we obtain the result

$$E[X|X_1 = 0] = \frac{n-2}{n-1}$$ ■

3.4.1 Computing Variances by Conditioning

Conditional expectations can also be used to compute the variance of a random variable. Specifically, we can use

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

and then use conditioning to obtain both $E[X]$ and $E[X^2]$. We illustrate this technique by determining the variance of a geometric random variable.

Example 3.19 (Variance of the Geometric Random Variable)**.** Independent trials, each resulting in a success with probability p, are performed in sequence. Let N be the trial number of the first success. Find $\text{Var}(N)$.

Solution: Let $Y = 1$ if the first trial results in a success, and $Y = 0$ otherwise.

$$\text{Var}(N) = E[N^2] - (E[N])^2$$

To calculate $E[N^2]$ and $E[N]$ we condition on Y. For instance,

$$E[N^2] = E\big[E[N^2|Y]\big]$$

However,

$$E[N^2|Y=1] = 1,$$
$$E[N^2|Y=0] = E[(1+N)^2]$$

These two equations are true since if the first trial results in a success, then clearly $N = 1$ and so $N^2 = 1$. On the other hand, if the first trial results in a failure, then the total number of trials necessary for the first success will equal one (the first trial that results in failure) plus the necessary number of additional trials. Since this latter quantity has the same distribution as N, we get that $E[N^2|Y=0] = E[(1+N)^2]$. Hence, we see that

$$\begin{aligned} E[N^2] &= E[N^2|Y=1]P\{Y=1\} + E[N^2|Y=0]P\{Y=0\} \\ &= p + E[(1+N)^2](1-p) \\ &= 1 + (1-p)E[2N + N^2] \end{aligned}$$

Since, as was shown in Example 3.11, $E[N] = 1/p$, this yields

$$E[N^2] = 1 + \frac{2(1-p)}{p} + (1-p)E[N^2]$$

or

$$E[N^2] = \frac{2-p}{p^2}$$

Therefore,

$$\begin{aligned} \mathrm{Var}(N) &= E[N^2] - (E[N])^2 \\ &= \frac{2-p}{p^2} - \left(\frac{1}{p}\right)^2 \\ &= \frac{1-p}{p^2} \end{aligned}$$

■

Another way to use conditioning to obtain the variance of a random variable is to apply the conditional variance formula. The conditional variance of X given that $Y = y$ is defined by

$$\mathrm{Var}(X|Y=y) = E\big[(X - E[X|Y=y])^2|Y=y\big]$$

That is, the conditional variance is defined in exactly the same manner as the ordinary variance with the exception that all probabilities are determined conditional on the event that $Y = y$. Expanding the right side of the preceding and taking expectation term by term yields

$$\mathrm{Var}(X|Y=y) = E[X^2|Y=y] - (E[X|Y=y])^2$$

Letting $\text{Var}(X|Y)$ denote that function of Y whose value when $Y = y$ is $\text{Var}(X|Y = y)$, we have the following result.

Proposition 3.1 (The Conditional Variance Formula)**.**

$$\text{Var}(X) = E\big[\text{Var}(X|Y)\big] + \text{Var}\big(E[X|Y]\big) \tag{3.7}$$

Proof.

$$\begin{aligned} E\big[\text{Var}(X|Y)\big] &= E[E[X^2|Y] - (E[X|Y])^2] \\ &= E\big[E[X^2|Y]\big] - E\big[(E[X|Y])^2\big] \\ &= E[X^2] - E\big[(E[X|Y])^2\big] \end{aligned}$$

and

$$\begin{aligned} \text{Var}(E[X|Y]) &= E\big[(E[X|Y])^2\big] - \big(E\big[E[X|Y]\big]\big)^2 \\ &= E\big[(E[X|Y])^2\big] - (E[X])^2 \end{aligned}$$

Therefore,

$$E\big[\text{Var}(X|Y)\big] + \text{Var}\big(E[X|Y]\big) = E[X^2] - (E[X])^2$$

which completes the proof. ■

Example 3.20 (The Variance of a Compound Random Variable)**.** Let $X_1, X_2, \ldots$ be independent and identically distributed random variables with distribution F having mean μ and variance σ^2, and assume that they are independent of the nonnegative integer valued random variable N. As noted in Example 3.10, where its expected value was determined, the random variable $S = \sum_{i=1}^{N} X_i$ is called a compound random variable. Find its variance.

Solution: Whereas we could obtain $E[S^2]$ by conditioning on N, let us instead use the conditional variance formula. Now,

$$\begin{aligned} \text{Var}(S|N = n) &= \text{Var}\left(\sum_{i=1}^{N} X_i | N = n\right) \\ &= \text{Var}\left(\sum_{i=1}^{n} X_i | N = n\right) \\ &= \text{Var}\left(\sum_{i=1}^{n} X_i\right) \\ &= n\sigma^2 \end{aligned}$$

By the same reasoning,

$$E[S|N = n] = n\mu$$

Therefore,

$$\text{Var}(S|N) = N\sigma^2, \quad E[S|N] = N\mu$$

and the conditional variance formula gives

$$\text{Var}(S) = E[N\sigma^2] + \text{Var}(N\mu) = \sigma^2 E[N] + \mu^2 \text{Var}(N)$$

If N is a Poisson random variable, then $S = \sum_{i=1}^{N} X_i$ is called a *compound Poisson* random variable. Because the variance of a Poisson random variable is equal to its mean, it follows that for a compound Poisson random variable having $E[N] = \lambda$

$$\text{Var}(S) = \lambda\sigma^2 + \lambda\mu^2 = \lambda E[X^2]$$

where X has the distribution F. ■

Example 3.21 (The Variance in the Matching Rounds Problem)**.** Consider the matching rounds problem of Example 3.14, and let $V_n = \text{Var}(R_n)$ denote the variance of the number of rounds needed when there are initially n people. Using the conditional variance formula, we will show that

$$V_n = n, \quad n \geq 2$$

The proof of the preceding is by induction on n. To begin, note that when $n = 2$ the number of rounds needed is geometric with parameter $p = 1/2$ and so

$$V_2 = \frac{1-p}{p^2} = 2$$

So assume the induction hypothesis that

$$V_j = j, \quad 2 \leq j < n$$

and now consider the case when there are n individuals. If X is the number of matches in the first round then, conditional on X, the number of rounds R_n is distributed as 1 plus the number of rounds needed when there are initially $n - X$ individuals. Consequently,

$$\begin{aligned} E[R_n|X] &= 1 + E[R_{n-X}] \\ &= 1 + n - X \qquad \text{by Example 3.14} \end{aligned}$$

Also, with $V_0 = 0$,

$$\text{Var}(R_n|X) = \text{Var}(R_{n-X}) = V_{n-X}$$

Hence, by the conditional variance formula

$$V_n = E[\text{Var}(R_n|X)] + \text{Var}(E[R_n|X])$$

$$
\begin{aligned}
&= E[V_{n-X}] + \mathrm{Var}(X) \\
&= \sum_{j=0}^{n} V_{n-j} P(X = j) + \mathrm{Var}(X) \\
&= V_n P(X = 0) + \sum_{j=1}^{n} V_{n-j} P(X = j) + \mathrm{Var}(X)
\end{aligned}
$$

Because $P(X = n-1) = 0$, it follows from the preceding and the induction hypothesis that

$$
\begin{aligned}
V_n &= V_n P(X = 0) + \sum_{j=1}^{n} (n - j) P(X = j) + \mathrm{Var}(X) \\
&= V_n P(X = 0) + n(1 - P(X = 0)) - E[X] + \mathrm{Var}(X)
\end{aligned}
$$

As it is easily shown (see Example 2.30 and Exercise 64 of Chapter 2) that $E[X] = \mathrm{Var}(X) = 1$, the preceding gives

$$
V_n = V_n P(X = 0) + n(1 - P(X = 0))
$$

thus proving the result. ■

3.5 Computing Probabilities by Conditioning

Not only can we obtain expectations by first conditioning on an appropriate random variable, but we may also use this approach to compute probabilities. To see this, let E denote an arbitrary event and define the indicator random variable X by

$$
X = \begin{cases} 1, & \text{if } E \text{ occurs} \\ 0, & \text{if } E \text{ does not occur} \end{cases}
$$

It follows from the definition of X that

$$
\begin{aligned}
E[X] &= P(E), \\
E[X|Y = y] &= P(E|Y = y), \quad \text{for any random variable } Y
\end{aligned}
$$

Therefore, from Eqs. (3.2a) and (3.2b) we obtain

$$
\begin{aligned}
P(E) &= \sum_{y} P(E|Y = y) P(Y = y), \quad \text{if } Y \text{ is discrete} \\
&= \int_{-\infty}^{\infty} P(E|Y = y) f_Y(y)\, dy, \quad \text{if } Y \text{ is continuous}
\end{aligned}
$$

Example 3.22. Suppose that X and Y are independent continuous random variables having densities f_X and f_Y, respectively. Compute $P\{X < Y\}$.

Solution: Conditioning on the value of Y yields

$$\begin{aligned} P\{X < Y\} &= \int_{-\infty}^{\infty} P\{X < Y | Y = y\} f_Y(y)\, dy \\ &= \int_{-\infty}^{\infty} P\{X < y | Y = y\} f_Y(y)\, dy \\ &= \int_{-\infty}^{\infty} P\{X < y\} f_Y(y)\, dy \\ &= \int_{-\infty}^{\infty} F_X(y) f_Y(y)\, dy \end{aligned}$$

where

$$F_X(y) = \int_{-\infty}^{y} f_X(x)\, dx$$ ■

Example 3.23. An insurance company supposes that the number of accidents that each of its policyholders will have in a year is Poisson distributed, with the mean of the Poisson depending on the policyholder. If the Poisson mean of a randomly chosen policyholder has a gamma distribution with density function

$$g(\lambda) = \lambda e^{-\lambda}, \quad \lambda \geq 0$$

what is the probability that a randomly chosen policyholder has exactly n accidents next year?

Solution: Let X denote the number of accidents that a randomly chosen policyholder has next year. Letting Y be the Poisson mean number of accidents for this policyholder, then conditioning on Y yields

$$\begin{aligned} P\{X = n\} &= \int_0^{\infty} P\{X = n | Y = \lambda\} g(\lambda)\, d\lambda \\ &= \int_0^{\infty} e^{-\lambda} \frac{\lambda^n}{n!} \lambda e^{-\lambda}\, d\lambda \\ &= \frac{1}{n!} \int_0^{\infty} \lambda^{n+1} e^{-2\lambda}\, d\lambda \end{aligned}$$

However, because

$$h(\lambda) = \frac{2e^{-2\lambda}(2\lambda)^{n+1}}{(n+1)!}, \quad \lambda > 0$$

is the density function of a gamma $(n+2, 2)$ random variable, its integral is 1. Therefore,

$$1 = \int_0^{\infty} \frac{2e^{-2\lambda}(2\lambda)^{n+1}}{(n+1)!}\, d\lambda = \frac{2^{n+2}}{(n+1)!} \int_0^{\infty} \lambda^{n+1} e^{-2\lambda}\, d\lambda$$

showing that

$$P\{X = n\} = \frac{n+1}{2^{n+2}} \qquad \blacksquare$$

Example 3.24. Suppose that the number of people who visit a yoga studio each day is a Poisson random variable with mean λ. Suppose further that each person who visits is, independently, female with probability p or male with probability $1 - p$. Find the joint probability that exactly n women and m men visit the academy today.

Solution: Let N_1 denote the number of women and N_2 the number of men who visit the academy today. Also, let $N = N_1 + N_2$ be the total number of people who visit. Conditioning on N gives

$$P\{N_1 = n, N_2 = m\} = \sum_{i=0}^{\infty} P\{N_1 = n, N_2 = m | N = i\} P\{N = i\}$$

Because $P\{N_1 = n, N_2 = m | N = i\} = 0$ when $i \neq n + m$, the preceding equation yields

$$P\{N_1 = n, N_2 = m\} = P\{N_1 = n, N_2 = m | N = n + m\} e^{-\lambda} \frac{\lambda^{n+m}}{(n+m)!}$$

Given that $n + m$ people visit it follows, because each of these $n + m$ is independently a woman with probability p, that the conditional probability that n of them are women (and m are men) is just the binomial probability of n successes in $n+m$ trials. Therefore,

$$\begin{aligned} P\{N_1 = n, N_2 = m\} &= \binom{n+m}{n} p^n (1-p)^m e^{-\lambda} \frac{\lambda^{n+m}}{(n+m)!} \\ &= \frac{(n+m)!}{n!m!} p^n (1-p)^m e^{-\lambda p} e^{-\lambda(1-p)} \frac{\lambda^n \lambda^m}{(n+m)!} \\ &= e^{-\lambda p} \frac{(\lambda p)^n}{n!} e^{-\lambda(1-p)} \frac{(\lambda(1-p))^m}{m!} \end{aligned}$$

Because the preceding joint probability mass function factors into two products, one of which depends only on n and the other only on m, it follows that N_1 and N_2 are independent. Moreover, because

$$\begin{aligned} P\{N_1 = n\} &= \sum_{m=0}^{\infty} P\{N_1 = n, N_2 = m\} \\ &= e^{-\lambda p} \frac{(\lambda p)^n}{n!} \sum_{m=0}^{\infty} e^{-\lambda(1-p)} \frac{(\lambda(1-p))^m}{m!} = e^{-\lambda p} \frac{(\lambda p)^n}{n!} \end{aligned}$$

and, similarly,

$$P\{N_2 = m\} = e^{-\lambda(1-p)} \frac{(\lambda(1-p))^m}{m!}$$

we can conclude that N_1 and N_2 are independent Poisson random variables with respective means λp and $\lambda(1-p)$. Therefore, this example establishes the important result that when each of a Poisson number of events is independently classified either as being type 1 with probability p or type 2 with probability $1-p$, then the numbers of type 1 and type 2 events are independent Poisson random variables. ■

The result of Example 3.24 generalizes to the case where each of a Poisson distributed number of events, N, with mean λ is independently classified as being one of k types, with the probability that it is type i being $p_i, i = 1, \ldots, k, \sum_{i=1}^{k} p_i = 1$. If N_i is the number that are classified as type i, then $N_1, \ldots, N_k$ are independent Poisson random variables with respective means $\lambda p_1, \ldots, \lambda p_k$. This follows, since for $n = \sum_{i=1}^{k} n_i$

$$\begin{aligned} P(N_1 = n_1, \ldots, N_k = n_k) &= P(N_1 = n_1, \ldots, N_k = n_k | N = n) P(N = n) \\ &= \frac{n!}{n_1! \cdots n_k!} p_1^{n_1} \cdots p_k^{n_k} e^{-\lambda} \lambda^n / n! \\ &= \prod_{i=1}^{k} e^{-\lambda p_i} (\lambda p_i)^{n_i} / n_i! \end{aligned}$$

where the second equality used that, given a total of n events, the numbers of each type has a multinomial distribution with parameters $(n, p_1, \ldots, p_k)$.

Example 3.25 (The Distribution of the Sum of Independent Bernoulli Random Variables)**.** Let $X_1, \ldots, X_n$ be independent Bernoulli random variables, with X_i having parameter $p_i, i = 1, \ldots, n$. That is, $P\{X_i = 1\} = p_i, P\{X_i = 0\} = q_i = 1 - p_i$. Suppose we want to compute the probability mass function of their sum, $X_1 + \cdots + X_n$. To do so, we will recursively obtain the probability mass function of $X_1 + \cdots + X_k$, first for $k = 1$, then $k = 2$, and on up to $k = n$. To begin, let

$$P_k(j) = P\{X_1 + \cdots + X_k = j\}$$

and note that

$$P_k(k) = \prod_{i=1}^{k} p_i, \qquad P_k(0) = \prod_{i=1}^{k} q_i$$

For $0 < j < k$, conditioning on X_k yields the recursion

$$\begin{aligned} P_k(j) &= P\{X_1 + \cdots + X_k = j | X_k = 1\} p_k + P\{X_1 + \cdots + X_k = j | X_k = 0\} q_k \\ &= P\{X_1 + \cdots + X_{k-1} = j - 1 | X_k = 1\} p_k \\ &\quad + P\{X_1 + \cdots + X_{k-1} = j | X_k = 0\} q_k \\ &= P\{X_1 + \cdots + X_{k-1} = j - 1\} p_k + P\{X_1 + \cdots + X_{k-1} = j\} q_k \\ &= p_k P_{k-1}(j-1) + q_k P_{k-1}(j) \end{aligned}$$

Starting with $P_1(1) = p_1$, $P_1(0) = q_1$, the preceding equations can be recursively solved to obtain the functions $P_2(j)$, $P_3(j)$, up to $P_n(j)$. ■

Example 3.26 (The Best Prize Problem)**.** Suppose that we are to be presented with n distinct prizes in sequence. After being presented with a prize we must immediately decide whether to accept it or reject it and consider the next prize. The only information we are given when deciding whether to accept a prize is the relative rank of that prize compared to ones already seen. That is, for instance, when the fifth prize is presented we learn how it compares with the first four prizes already seen. Suppose that once a prize is rejected it is lost, and that our objective is to maximize the probability of obtaining the best prize. Assuming that all $n!$ orderings of the prizes are equally likely, how well can we do?

Solution: Rather surprisingly, we can do quite well. To see this, fix a value k, $0 \leq k < n$, and consider the strategy that rejects the first k prizes and then accepts the first one that is better than all of those first k. Let P_k (best) denote the probability that the best prize is selected when this strategy is employed. To compute this probability, condition on X, the position of the best prize. This gives

$$\begin{aligned} P_k(\text{best}) &= \sum_{i=1}^{n} P_k(\text{best}|X=i)P(X=i) \\ &= \frac{1}{n}\sum_{i=1}^{n} P_k(\text{best}|X=i) \end{aligned}$$

Now, if the overall best prize is among the first k, then no prize is ever selected under the strategy considered. On the other hand, if the best prize is in position i, where $i > k$, then the best prize will be selected if the best of the first k prizes is also the best of the first $i-1$ prizes (for then none of the prizes in positions $k+1, k+2, \ldots, i-1$ would be selected). Hence, we see that

$$\begin{aligned} P_k(\text{best}|X=i) &= 0, \quad \text{if } i \leq k \\ P_k(\text{best}|X=i) &= P\{\text{best of first } i-1 \text{ is among the first } k\} \\ &= k/(i-1), \quad \text{if } i > k \end{aligned}$$

From the preceding, we obtain

$$\begin{aligned} P_k(\text{best}) &= \frac{k}{n}\sum_{i=k+1}^{n}\frac{1}{i-1} \\ &\approx \frac{k}{n}\int_{k}^{n-1}\frac{1}{x}\,dx \\ &= \frac{k}{n}\log\left(\frac{n-1}{k}\right) \\ &\approx \frac{k}{n}\log\left(\frac{n}{k}\right) \end{aligned}$$

Now, if we consider the function

$$g(x) = \frac{x}{n} \log\left(\frac{n}{x}\right)$$

then

$$g'(x) = \frac{1}{n} \log\left(\frac{n}{x}\right) - \frac{1}{n}$$

and so

$$g'(x) = 0 \Rightarrow \log(n/x) = 1 \Rightarrow x = n/e$$

Thus, since $P_k(\text{best}) \approx g(k)$, we see that the best strategy of the type considered is to let the first n/e prizes go by and then accept the first one to appear that is better than all of those. In addition, since $g(n/e) = 1/e$, the probability that this strategy selects the best prize is approximately $1/e \approx 0.36788$.

Remark. Most students are quite surprised by the size of the probability of obtaining the best prize, thinking that this probability would be close to 0 when n is large. However, even without going through the calculations, a little thought reveals that the probability of obtaining the best prize can be made to be reasonably large. Consider the strategy of letting half of the prizes go by, and then selecting the first one to appear that is better than all of those. The probability that a prize is actually selected is the probability that the overall best is among the second half and this is $1/2$. In addition, given that a prize is selected, at the time of selection that prize would have been the best of more than $n/2$ prizes to have appeared, and would thus have probability of at least $1/2$ of being the overall best. Hence, the strategy of letting the first half of all prizes go by and then accepting the first one that is better than all of those prizes results in a probability greater than $1/4$ of obtaining the best prize. ■

Example 3.27. At a party n men take off their hats. The hats are then mixed up and each man randomly selects one. We say that a match occurs if a man selects his own hat. What is the probability of no matches? What is the probability of exactly k matches?

Solution: Let E denote the event that no matches occur, and to make explicit the dependence on n, write $P_n = P(E)$. We start by conditioning on whether or not the first man selects his own hat—call these events M and M^c. Then

$$P_n = P(E) = P(E|M)P(M) + P(E|M^c)P(M^c)$$

Clearly, $P(E|M) = 0$, and so

$$P_n = P(E|M^c)\frac{n-1}{n} \tag{3.8}$$

Now, $P(E|M^c)$ is the probability of no matches when $n-1$ men select from a set of $n-1$ hats that does not contain the hat of one of these men. This can happen

in either of two mutually exclusive ways. Either there are no matches and the extra man does not select the extra hat (this being the hat of the man that chose first), or there are no matches and the extra man does select the extra hat. The probability of the first of these events is just P_{n-1}, which is seen by regarding the extra hat as "belonging" to the extra man. Because the second event has probability $[1/(n-1)]P_{n-2}$, we have

$$P(E|M^c) = P_{n-1} + \frac{1}{n-1}P_{n-2}$$

and thus, from Eq. (3.8),

$$P_n = \frac{n-1}{n}P_{n-1} + \frac{1}{n}P_{n-2}$$

or, equivalently,

$$P_n - P_{n-1} = -\frac{1}{n}(P_{n-1} - P_{n-2}) \tag{3.9}$$

However, because P_n is the probability of no matches when n men select among their own hats, we have

$$P_1 = 0, \quad P_2 = \tfrac{1}{2}$$

and so, from Eq. (3.9),

$$P_3 - P_2 = -\frac{(P_2 - P_1)}{3} = -\frac{1}{3!} \quad \text{or} \quad P_3 = \frac{1}{2!} - \frac{1}{3!},$$
$$P_4 - P_3 = -\frac{(P_3 - P_2)}{4} = \frac{1}{4!} \quad \text{or} \quad P_4 = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}$$

and, in general, we see that

$$P_n = \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots + \frac{(-1)^n}{n!}$$

To obtain the probability of exactly k matches, we consider any fixed group of k men. The probability that they, and only they, select their own hats is

$$\frac{1}{n}\frac{1}{n-1}\cdots\frac{1}{n-(k-1)}P_{n-k} = \frac{(n-k)!}{n!}P_{n-k}$$

where P_{n-k} is the conditional probability that the other $n-k$ men, selecting among their own hats, have no matches. Because there are $\binom{n}{k}$ choices of a set of k men, the desired probability of exactly k matches is

$$\frac{P_{n-k}}{k!} = \frac{\frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^{n-k}}{(n-k)!}}{k!}$$

which, for n large, is approximately equal to $e^{-1}/k!$.

Remark. The recursive equation, Eq. (3.9), could also have been obtained by using the concept of a cycle, where we say that the sequence of distinct individuals $i_1, i_2, \ldots, i_k$ constitutes a *cycle* if i_1 chooses i_2's hat, i_2 chooses i_3's hat, $\ldots$, i_{k-1} chooses i_k's hat, and i_k chooses i_1's hat. Note that every individual is part of a cycle, and that a cycle of size $k = 1$ occurs when someone chooses his or her own hat. With E being, as before, the event that no matches occur, it follows upon conditioning on the size of the cycle containing a specified person, say person 1, that

$$P_n = P(E) = \sum_{k=1}^{n} P(E|C = k)P(C = k) \tag{3.10}$$

where C is the size of the cycle that contains person 1. Now, call person 1 the first person, and note that $C = k$ if the first person does not choose 1's hat; the person whose hat was chosen by the first person—call this person the second person—does not choose 1's hat; the person whose hat was chosen by the second person—call this person the third person—does not choose 1's hat; $\ldots$, the person whose hat was chosen by the $(k-1)$st person does choose 1's hat. Consequently,

$$P(C = k) = \frac{n-1}{n}\frac{n-2}{n-1}\cdots\frac{n-k+1}{n-k+2}\frac{1}{n-k+1} = \frac{1}{n} \tag{3.11}$$

That is, the size of the cycle that contains a specified person is equally likely to be any of the values $1, 2, \ldots, n$. Moreover, since $C = 1$ means that 1 chooses his or her own hat, it follows that

$$P(E|C = 1) = 0$$

On the other hand, if $C = k$, then the set of hats chosen by the k individuals in this cycle is exactly the set of hats of these individuals. Hence, conditional on $C = k$, the problem reduces to determining the probability of no matches when $n - k$ people randomly choose among their own $n - k$ hats. Therefore, for $k > 1$

$$P(E|C = k) = P_{n-k} \tag{3.12}$$

Substituting (3.11)–(3.13) back into Eq. (3.10) gives

$$P_n = \frac{1}{n}\sum_{k=2}^{n} P_{n-k} \tag{3.13}$$

which is easily shown to be equivalent to Eq. (3.9). ■

Example 3.28 (The Ballot Problem)**.** In an election, candidate A receives n votes, and candidate B receives m votes where $n > m$. Assuming that all orderings are equally likely, show that the probability that A is always ahead in the count of votes is $(n - m)/(n + m)$.

Solution: Let $P_{n,m}$ denote the desired probability. By conditioning on which candidate receives the last vote counted we have

$$\begin{aligned} P_{n,m} &= P\{A \text{ always ahead} | A \text{ receives last vote}\}\frac{n}{n+m} \\ &\quad + P\{A \text{ always ahead} | B \text{ receives last vote}\}\frac{m}{n+m} \end{aligned}$$

Now, given that A receives the last vote, we can see that the probability that A is always ahead is the same as if A had received a total of $n - 1$ and B a total of m votes. Because a similar result is true when we are given that B receives the last vote, we see from the preceding that

$$P_{n,m} = \frac{n}{n+m}P_{n-1,m} + \frac{m}{m+n}P_{n,m-1} \tag{3.14}$$

We can now prove that $P_{n,m} = (n - m)/(n + m)$ by induction on $n + m$. As it is true when $n + m = 1$, that is, $P_{1,0} = 1$, assume it whenever $n + m = k$. Then when $n + m = k + 1$, we have by Eq. (3.14) and the induction hypothesis that

$$\begin{aligned} P_{n,m} &= \frac{n}{n+m}\frac{n-1-m}{n-1+m} + \frac{m}{m+n}\frac{n-m+1}{n+m-1} \\ &= \frac{n-m}{n+m} \end{aligned}$$

and the result is proven. ■

The ballot problem has some interesting applications. For example, consider successive flips of a coin that always land on "heads" with probability p, and let us determine the probability distribution of the first time, after beginning, that the total number of heads is equal to the total number of tails. The probability that the first time this occurs is at time $2n$ can be obtained by first conditioning on the total number of heads in the first $2n$ trials. This yields

$$\begin{aligned} &P\{\text{first time equal} = 2n\} \\ &\quad = P\{\text{first time equal} = 2n | n \text{ heads in first } 2n\}\binom{2n}{n}p^n(1-p)^n \end{aligned}$$

Now, given a total of n heads in the first $2n$ flips we can see that all possible orderings of the n heads and n tails are equally likely, and thus the preceding conditional probability is equivalent to the probability that in an election, in which each candidate receives n votes, one of the candidates is always ahead in the counting until the last

vote (which ties them). But by conditioning on whomever receives the last vote, we see that this is just the probability in the ballot problem when $m = n - 1$. Hence,

$$P\{\text{first time equal} = 2n\} = P_{n,n-1}\binom{2n}{n}p^n(1-p)^n$$
$$= \frac{\binom{2n}{n}p^n(1-p)^n}{2n-1}$$

Suppose now that we wanted to determine the probability that the first time there are i more heads than tails occurs after the $(2n+i)$th flip. Now, in order for this to be the case, the following two events must occur:

(a) The first $2n+i$ tosses result in $n+i$ heads and n tails; and
(b) The order in which the $n+i$ heads and n tails occur is such that the number of heads is never i more than the number of tails until after the final flip.

Now, it is easy to see that event (b) will occur if and only if the order of appearance of the $n+i$ heads and n tails is such that starting from the final flip and working backwards heads is always in the lead. For instance, if there are 4 heads and 2 tails $(n = 2, i = 2)$, then the outcome $____TH$ would not suffice because there would have been 2 more heads than tails sometime before the sixth flip (since the first 4 flips resulted in 2 more heads than tails).

Now, the probability of the event specified in (a) is just the binomial probability of getting $n+i$ heads and n tails in $2n+i$ flips of the coin.

We must now determine the conditional probability of the event specified in (b) given that there are $n+i$ heads and n tails in the first $2n+i$ flips. To do so, note first that given that there are a total of $n+i$ heads and n tails in the first $2n+i$ flips, all possible orderings of these flips are equally likely. As a result, the conditional probability of (b) given (a) is just the probability that a random ordering of $n+i$ heads and n tails will, when counted in reverse order, always have more heads than tails. Since all reverse orderings are also equally likely, it follows from the ballot problem that this conditional probability is $i/(2n+i)$.

That is, we have shown that

$$P\{a\} = \binom{2n+i}{n}p^{n+i}(1-p)^n,$$
$$P\{b|a\} = \frac{i}{2n+i}$$

and so

$$P\{\text{first time heads leads by } i \text{ is after flip } 2n+i\} = \binom{2n+i}{n}p^{n+i}(1-p)^n\frac{i}{2n+i}$$

Example 3.29. Let $U_1, U_2, \ldots$ be a sequence of independent uniform (0, 1) random variables, and let

$$N = \min\{n \geq 2: U_n > U_{n-1}\}$$

and

$$M = \min\{n \geq 1: U_1 + \cdots + U_n > 1\}$$

That is, N is the index of the first uniform random variable that is larger than its immediate predecessor, and M is the number of uniform random variables we need sum to exceed 1. Surprisingly, N and M have the same probability distribution, and their common mean is e!

Solution: It is easy to find the distribution of N. Since all $n!$ possible orderings of $U_1, \ldots, U_n$ are equally likely, we have

$$P\{N > n\} = P\{U_1 > U_2 > \cdots > U_n\} = 1/n!$$

To show that $P\{M > n\} = 1/n!$, we will use mathematical induction. However, to give ourselves a stronger result to use as the induction hypothesis, we will prove the stronger result that for $0 < x \leq 1$, $P\{M(x) > n\} = x^n/n!, n \geq 1$, where

$$M(x) = \min\{n \geq 1: U_1 + \cdots + U_n > x\}$$

is the minimum number of uniforms that need be summed to exceed x. To prove that $P\{M(x) > n\} = x^n/n!$, note first that it is true for $n = 1$ since

$$P\{M(x) > 1\} = P\{U_1 \leq x\} = x$$

So assume that for all $0 < x \leq 1$, $P\{M(x) > n\} = x^n/n!$. To determine $P\{M(x) > n + 1\}$, condition on U_1 to obtain

$$\begin{aligned}
P\{M(x) > n+1\} &= \int_0^1 P\{M(x) > n+1 | U_1 = y\}\, dy \\
&= \int_0^x P\{M(x) > n+1 | U_1 = y\}\, dy \\
&= \int_0^x P\{M(x-y) > n\}\, dy \\
&= \int_0^x \frac{(x-y)^n}{n!}\, dy \quad \text{by the induction hypothesis} \\
&= \int_0^x \frac{u^n}{n!}\, du \\
&= \frac{x^{n+1}}{(n+1)!}
\end{aligned}$$

where the third equality of the preceding follows from the fact that given $U_1 = y$, $M(x)$ is distributed as 1 plus the number of uniforms that need be summed to exceed $x - y$. Thus, the induction is complete and we have shown that for $0 < x \leq 1, n \geq 1$,

$$P\{M(x) > n\} = x^n/n!$$

Letting $x = 1$ shows that N and M have the same distribution. Finally, we have

$$E[M] = E[N] = \sum_{n=0}^{\infty} P\{N > n\} = \sum_{n=0}^{\infty} 1/n! = e \qquad \blacksquare$$

Example 3.30. Let $X_1, X_2, \ldots$ be independent continuous random variables with a common distribution function F and density $f = F'$, and suppose that they are to be observed one at a time in sequence. Let

$$N = \min\{n \geqslant 2\text{: } X_n = \text{second largest of } X_1, \ldots, X_n\}$$

and let

$$M = \min\{n \geq 2\text{: } X_n = \text{second smallest of } X_1, \ldots, X_n\}$$

Which random variable—X_N, the first random variable which when observed is the second largest of those that have been seen, or X_M, the first one that on observation is the second smallest to have been seen—tends to be larger?

Solution: To calculate the probability density function of X_N, it is natural to condition on the value of N; so let us start by determining its probability mass function. Now, if we let

$$A_i = \{X_i \neq \text{second largest of } X_1, \ldots, X_i\}, \quad i \geq 2$$

then, for $n \geq 2$,

$$P\{N = n\} = P\left(A_2 A_3 \cdots A_{n-1} A_n^c\right)$$

Since the X_i are independent and identically distributed it follows that, for any $m \geq 1$, knowing the rank ordering of the variables $X_1, \ldots, X_m$ yields no information about the set of m values $\{X_1, \ldots, X_m\}$. That is, for instance, knowing that $X_1 < X_2$ gives us no information about the values of $\min(X_1, X_2)$ or $\max(X_1, X_2)$. It follows from this that the events $A_i, i \geq 2$ are independent. Also, since X_i is equally likely to be the largest, or the second largest, $\ldots$, or the ith largest of $X_1, \ldots, X_i$ it follows that $P(A_i) = (i-1)/i, i \geq 2$. Therefore, we see that

$$P\{N = n\} = \frac{1}{2}\frac{2}{3}\frac{3}{4} \cdots \frac{n-2}{n-1}\frac{1}{n} = \frac{1}{n(n-1)}$$

Hence, conditioning on N yields that the probability density function of X_N is as follows:

$$f_{X_N}(x) = \sum_{n=2}^{\infty} \frac{1}{n(n-1)} f_{X_N|N}(x|n)$$

Now, since the ordering of the variables $X_1, \ldots, X_n$ is independent of the set of values $\{X_1, \ldots, X_n\}$, it follows that the event $\{N = n\}$ is independent of

$\{X_1, \ldots, X_n\}$. From this, it follows that the conditional distribution of X_N given that $N = n$ is equal to the distribution of the second largest from a set of n random variables having distribution F. Thus, using the results of Example 2.38 concerning the density function of such a random variable, we obtain

$$\begin{aligned} f_{X_N}(x) &= \sum_{n=2}^{\infty} \frac{1}{n(n-1)} \frac{n!}{(n-2)!1!} (F(x))^{n-2} f(x)(1 - F(x)) \\ &= f(x)(1 - F(x)) \sum_{i=0}^{\infty} (F(x))^i \\ &= f(x) \end{aligned}$$

Thus, rather surprisingly, X_N has the same distribution as X_1, namely, F. Also, if we now let $W_i = -X_i$, $i \geq 1$, then W_M will be the value of the first W_i, which on observation is the second largest of all those that have been seen. Hence, by the preceding, it follows that W_M has the same distribution as W_1. That is, $-X_M$ has the same distribution as $-X_1$, and so X_M also has distribution F! In other words, whether we stop at the first random variable that is the second largest of all those presently observed, or we stop at the first one that is the second smallest of all those presently observed, we will end up with a random variable having distribution F.

Whereas the preceding result is quite surprising, it is a special case of a general result known as *Ignatov's theorem*, which yields even more surprises. For instance, for $k \geqslant 1$, let

$$N_k = \min\{n \geqslant k\colon X_n = k\text{th largest of } X_1, \ldots, X_n\}$$

Therefore, N_2 is what we previously called N, and X_{N_k} is the first random variable that upon observation is the kth largest of all those observed up to this point. It can then be shown by a similar argument as used in the preceding that X_{N_k} has distribution function F for all k (see Exercise 82 at the end of this chapter). In addition, it can be shown that the random variables $X_{N_k}, k \geqslant 1$ are independent. (A statement and proof of Ignatov's theorem in the case of discrete random variables are given in Section 3.6.6.) ■

Example 3.31. A population consists of m families. Let X_j denote the size of family j, and suppose that $X_1, \ldots, X_m$ are independent random variables having the common probability mass function

$$p_k = P(X_j = k), \quad \sum_{k=1}^{\infty} p_k = 1$$

with mean $\mu = \Sigma_k k p_k$. Suppose a member of the population is randomly chosen, in that the selection is equally likely to be any of the members of the population, and let

S_i be the event that the selected individual is from a family of size i. Argue that

$$P(S_i) \to \frac{ip_i}{\mu} \text{ as } \quad m \to \infty$$

Solution: A heuristic argument for the preceding formula is that because each family is of size i with probability p_i, it follows that there are approximately mp_i families of size i when m is large. Thus, imp_i members of the population come from a family of size i, implying that the probability that the selected individual is from a family of size i is approximately $\frac{imp_i}{\sum_j jmp_j} = \frac{ip_i}{\mu}$.

For a more formal argument, let N_i denote the number of families that are of size i. That is,

$$N_i = \text{number } \{k : k = 1, \ldots, m : X_k = i\}$$

Then, conditional on $\mathbf{X} = (X_1, \ldots, X_m)$

$$P(S_i|\mathbf{X}) = \frac{iN_i}{\sum_{k=1}^m X_k}$$

Hence,

$$\begin{aligned} P(S_i) &= E[P(S_i|X)] \\ &= E\left[\frac{iN_i}{\sum_{k=1}^m X_k}\right] \\ &= E\left[\frac{iN_i/m}{\sum_{k=1}^m X_k/m}\right] \end{aligned}$$

Because each family is independently of size i with probability p_i, it follows by the strong law of large numbers that N_i/m, the fraction of families that are of size i, converges to p_i as $m \to \infty$. Also by the strong law of large numbers, $\sum_{k=1}^m X_k/m \to E[X] = \mu$ as $m \to \infty$. Consequently, with probability 1,

$$\frac{iN_i/m}{\sum_{k=1}^m X_k/m} \to \frac{ip_i}{\mu} \quad \text{as } m \to \infty$$

Because the random variable $\frac{iN_i}{\sum_{k=1}^m X_k}$ converges to $\frac{ip_i}{\mu}$ so does its expectation, which proves the result. (While it is not always the case that $\lim_{m\to\infty} Y_m = c$ implies that $\lim_{m\to\infty} E[Y_m] = c$, the implication is true when the Y_m are uniformly bounded random variables, and the random variables $\frac{iNi}{\sum_{k=1}^m X_k}$ are all between 0 and 1.) ■

The use of conditioning can also result in a more computationally efficient solution than a direct calculation. This is illustrated by our next example.

Example 3.32. Consider n independent trials in which each trial results in one of the outcomes $1, \ldots, k$ with respective probabilities $p_1, \ldots, p_k$, $\sum_{i=1}^k p_i = 1$. Suppose

further that $n > k$, and that we are interested in determining the probability that each outcome occurs at least once. If we let A_i denote the event that outcome i does not occur in any of the n trials, then the desired probability is $1 - P(\bigcup_{i=1}^{k} A_i)$, and it can be obtained by using the inclusion–exclusion theorem as follows:

$$P\left(\bigcup_{i=1}^{k} A_i\right) = \sum_{i=1}^{k} P(A_i) - \sum_{i}\sum_{j>i} P(A_i A_j) + \sum_{i}\sum_{j>i}\sum_{k>j} P(A_i A_j A_k) - \cdots + (-1)^{k+1} P(A_1 \cdots A_k)$$

where

$$\begin{aligned} P(A_i) &= (1 - p_i)^n \\ P(A_i A_j) &= (1 - p_i - p_j)^n, \quad i < j \\ P(A_i A_j A_k) &= (1 - p_i - p_j - p_k)^n, \quad i < j < k \end{aligned}$$

The difficulty with the preceding solution is that its computation requires the calculation of $2^k - 1$ terms, each of which is a quantity raised to the power n. The preceding solution is thus computationally inefficient when k is large. Let us now see how to make use of conditioning to obtain an efficient solution.

To begin, note that if we start by conditioning on N_k (the number of times that outcome k occurs) then when $N_k > 0$ the resulting conditional probability will equal the probability that all of the outcomes $1, \ldots, k-1$ occur at least once when $n - N_k$ trials are performed, and each results in outcome i with probability $p_i / \sum_{j=1}^{k-1} p_j$, $i = 1, \ldots, k-1$. We could then use a similar conditioning step on these terms.

To follow through on the preceding idea, let $A_{m,r}$, for $m \leqslant n, r \leqslant k$, denote the event that each of the outcomes $1, \ldots, r$ occurs at least once when m independent trials are performed, where each trial results in one of the outcomes $1, \ldots, r$ with respective probabilities $p_1/P_r, \ldots, p_r/P_r$, where $P_r = \sum_{j=1}^{r} p_j$. Let $P(m, r) = P(A_{m,r})$ and note that $P(n, k)$ is the desired probability. To obtain an expression for $P(m, r)$, condition on the number of times that outcome r occurs. This gives

$$\begin{aligned} P(m, r) &= \sum_{j=0}^{m} P\{A_{m,r} | r \text{ occurs } j \text{ times}\} \binom{m}{j} \left(\frac{p_r}{P_r}\right)^j \left(1 - \frac{p_r}{P_r}\right)^{m-j} \\ &= \sum_{j=1}^{m-r+1} P(m - j, r - 1) \binom{m}{j} \left(\frac{p_r}{P_r}\right)^j \left(1 - \frac{p_r}{P_r}\right)^{m-j} \end{aligned}$$

Starting with

$$\begin{aligned} P(m, 1) &= 1, \quad \text{if } m \geqslant 1 \\ P(m, 1) &= 0, \quad \text{if } m = 0 \end{aligned}$$

we can use the preceding recursion to obtain the quantities $P(m, 2), m = 2, \ldots, n - (k - 2)$, and then the quantities $P(m, 3), m = 3, \ldots, n - (k - 3)$, and so on, up to $P(m, k - 1)$, $m = k - 1, \ldots, n - 1$. At this point we can then use the recursion to compute $P(n, k)$. It is not difficult to check that the amount of computation needed is a polynomial function of k, which will be much smaller than 2^k when k is large. ■

Our next example is concerned with final score probabilities in serve and rally games such as table tennis, squash, paddle ball, volleyball, and others.

Example 3.33 (Serve and Rally Competitions)**.** Consider a serve and rally competition involving players A and B. Suppose that each rally that begins with a serve by player A is won by player A with probability p_a and is won by player B with probability $q_a = 1 - p_a$. Furthermore, suppose that each rally that begins with a serve by player B is won by player A with probability p_b and is won by player B with probability $q_b = 1 - p_b$. Suppose that the winner of the rally earns a point and becomes the server of the next rally. The competition is decided either when A has won a total of N points or when B has won a total of M. Given that A serves first, we are interested in determining the final score probabilities.

The format of this example is used in a variety of serve and rally games, including international volleyball and American squash, both of which changed from their original format which gave service to the winner of the previous rally but only awarded a point if the winner of a rally was the server. (See Exercise 84 for an analysis of this latter format.)

Let F denote the final score, with $F = (i, j)$ meaning that A won a total of i points and B a total of j points. Clearly

$$P(F = (N, 0)) = p_a^N, \quad P(F = (0, M)) = q_a q_b^{M-1}$$

To determine the other final score probabilities, imagine that A and B continue to play even after the competition is decided. Define the concept of a "round" by letting the initial serve of A start the first round and letting a new round begin each time A serves. Let B_i denote the number of points won by B in round i. Note that if the first point of a round is won by A, then that round ends with B winning 0 points in it. On the other hand, if B wins the first point in a round then B will continue serving until A wins a point, showing that the number of points won by B in a round is equal to the number of times that B serves. Because the number of consecutive serves of B before A wins a point is geometric with parameter p_b, we see that

$$B_i = \begin{cases} 0, & \text{with probability } p_a \\ \text{Geometric}(p_b), & \text{with probability } q_a \end{cases}$$

That is,

$$P(B_i = 0) = p_a$$
$$P(B_i = k | B_i > 0) = q_b^{k-1} p_b, k > 0$$

Because a new round begins each time A wins a point, it follows that B_i is the number of points that B wins between the time that A has won $i-1$ points until A has won i points. Consequently, $B(n) \equiv \sum_{i=1}^{n} B_i$ is the number of points that B has won at the moment that A wins its nth point. Noting that the final score will be $(N, m), m < M$, if $B(N) = m$, let us determine $P(B(n) = m)$ for $m > 0$. To do so, we condition on the number of $B_1, \ldots, B_n$ that are positive. Calling this number Y, that is,

$$Y = \text{number of } i \leq n \text{ such that } B_i > 0$$

we obtain

$$\begin{aligned} P(B(n) = m) &= \sum_{r=0}^{n} P(B(n) = m | Y = r) P(Y = r) \\ &= \sum_{r=1}^{n} P(B(n) = m | Y = r) P(Y = r) \end{aligned}$$

where the last equality followed since $m > 0$ and so $P(B(n) = m | Y = 0) = 0$. Because each of $B_1, \ldots, B_n$ is independently positive with probability q_a, it follows that Y, the number of them that are positive, is binomial with parameters n, q_a. Consequently,

$$P(B(n) = m) = \sum_{r=1}^{n} P(B(n) = m | Y = r) \binom{n}{r} q_a^r p_a^{n-r}$$

Now, if r of the variables $B_1, \ldots, B_n$ are positive, then $B(n)$ is distributed as the sum of r independent geometric random variables with parameter p_b, which is the negative binomial distribution of the number of trials until there have been r successes when each trial is independently a success with probability p_b. Hence,

$$P(B(n) = m | Y = r) = \binom{m-1}{r-1} p_b^r q_b^{m-r}$$

where we are using the convention that $\binom{a}{b} = 0$ if $b > a$. This gives

$$\begin{aligned} P(B(n) = m) &= \sum_{r=1}^{n} \binom{m-1}{r-1} p_b^r q_b^{m-r} \binom{n}{r} q_a^r p_a^{n-r} \\ &= q_b^m p_a^n \sum_{r=1}^{n} \binom{m-1}{r-1} \binom{n}{r} \left(\frac{p_b q_a}{q_b p_a} \right)^r \end{aligned}$$

Thus, we have shown that

$$P(F = (N, m)) = P(B(N) = m)$$

$$= q_b^m p_a^N \sum_{r=1}^{N} \binom{m-1}{r-1} \binom{N}{r} \left(\frac{p_b q_a}{q_b p_a}\right)^r, \quad 0 < m < M$$

To determine the probability that the final score will be $(n, M), 0 < n < N$, we condition on the number of wins that B has at the moment that A wins its nth game to obtain

$$\begin{aligned} P(F = (n, M)) &= \sum_{m=0}^{\infty} P(F = (n, M) | B(n) = m) P(B(n) = m) \\ &= \sum_{m=0}^{M-1} P(F = (n, M) | B(n) = m) P(B(n) = m) \end{aligned}$$

Now, given that B has $m < M$ points at the moment that A wins its nth point, in order for the final score to be (n, M) B must win the next point with A serving and must then win the final $M - m - 1$ points on its serve. Hence, $P(F = (n, M) | B(n) = m) = q_a q_b^{M-m-1}$, giving that

$$\begin{aligned} P(F = (n, M)) &= \sum_{m=0}^{M-1} q_a q_b^{M-m-1} P(B(n) = m) \\ &= q_a q_b^{M-1} p_a^n + \sum_{m=1}^{M-1} q_a q_b^{M-m-1} P(B(n) = m) \\ &= q_a q_b^{M-1} p_a^n \left[1 + \sum_{m=1}^{M-1} \sum_{r=1}^{n} \binom{m-1}{r-1} \binom{n}{r} \left(\frac{p_b q_a}{q_b p_a}\right)^r \right], \\ & \qquad\qquad 0 < n < N \end{aligned}$$

■

As noted previously, conditional expectations given that $Y = y$ are exactly the same as ordinary expectations except that all probabilities are computed conditional on the event that $Y = y$. As such, conditional expectations satisfy all the properties of ordinary expectations. For instance, the analog of

$$E[X] = \begin{cases} \sum_w E[X | W = w] P\{W = w\}, & \text{if } W \text{ is discrete} \\ \int_w E[X | W = w] f_W(w)\, dw, & \text{if } W \text{ is continuous} \end{cases}$$

is

$$E[X | Y = y] = \begin{cases} \sum_w E[X | W = w, Y = y] P\{W = w | Y = y\}, & \text{if } W \text{ is discrete} \\ \int_w E[X | W = w, Y = y] f_{W|Y}(w|y)\, dw, & \text{if } W \text{ is continuous} \end{cases}$$

If $E[X|Y, W]$ is defined to be that function of Y and W that, when $Y = y$, and $W = w$, is equal to $E[X|Y = y, W = w]$, then the preceding can be written as

$$E[X|Y] = E\big[E[X|Y, W]|Y\big]$$

Example 3.34. An automobile insurance company classifies each of its policyholders as being of one of the types $i = 1, \ldots, k$. It supposes that the numbers of accidents that a type i policyholder has in successive years are independent Poisson random variables with mean λ_i, $i = 1, \ldots, k$. The probability that a newly insured policyholder is type i is p_i, $\sum_{i=1}^{k} p_i = 1$. Given that a policyholder had n accidents in her first year, what is the expected number that she has in her second year? What is the conditional probability that she has m accidents in her second year?

Solution: Let N_i denote the number of accidents the policyholder has in year $i, i = 1, 2$. To obtain $E[N_2|N_1 = n]$, condition on her risk type T.

$$\begin{aligned}
E[N_2|N_1 = n] &= \sum_{j=1}^{k} E[N_2|T = j, N_1 = n]P\{T = j|N_1 = n\} \\
&= \sum_{j=1}^{k} E[N_2|T = j]P\{T = j|N_1 = n\} \\
&= \sum_{j=1}^{k} \lambda_j P\{T = j|N_1 = n\} \\
&= \frac{\sum_{j=1}^{k} e^{-\lambda_j}\lambda_j^{n+1} p_j}{\sum_{j=1}^{k} e^{-\lambda_j}\lambda_j^{n} p_j}
\end{aligned}$$

where the final equality used that

$$\begin{aligned}
P\{T = j|N_1 = n\} &= \frac{P\{T = j, N_1 = n\}}{P\{N_1 = n\}} \\
&= \frac{P\{N_1 = n|T = j\}P\{T = j\}}{\sum_{j=1}^{k} P\{N_1 = n|T = j\}P\{T = j\}} \\
&= \frac{p_j e^{-\lambda_j}\lambda_j^n/n!}{\sum_{j=1}^{k} p_j e^{-\lambda_j}\lambda_j^n/n!}
\end{aligned}$$

The conditional probability that the policyholder has m accidents in year 2 given that she had n in year 1 can also be obtained by conditioning on her type.

$$\begin{aligned}
P\{N_2 = m|N_1 = n\} &= \sum_{j=1}^{k} P\{N_2 = m|T = j, N_1 = n\}P\{T = j|N_1 = n\} \\
&= \sum_{j=1}^{k} e^{-\lambda_j}\frac{\lambda_j^m}{m!} P\{T = j|N_1 = n\}
\end{aligned}$$

Thus,

$$
\begin{aligned}
P_r(1) &= \frac{r(1-p)}{p}\alpha_1 P_{r+1}(0) \\
&= rp^r(1-p)\alpha_1, \\
P_r(2) &= \frac{r(1-p)}{2p}\big[\alpha_1 P_{r+1}(1) + 2\alpha_2 P_{r+1}(0)\big] \\
&= \frac{r(1-p)}{2p}\big[\alpha_1^2(r+1)p^{r+1}(1-p) + 2\alpha_2 p^{r+1}\big], \\
P_r(3) &= \frac{r(1-p)}{3p}\big[\alpha_1 P_{r+1}(2) + 2\alpha_2 P_{r+1}(1) + 3\alpha_3 P_{r+1}(0)\big]
\end{aligned}
$$

and so on.

Exercises

1. If X and Y are both discrete, show that $\sum_x p_{X|Y}(x|y) = 1$ for all y such that $p_Y(y) > 0$.

***2.** Let X_1 and X_2 be independent geometric random variables having the same parameter p. Guess the value of

$$P\{X_1 = i | X_1 + X_2 = n\}$$

Hint: Suppose a coin having probability p of coming up heads is continually flipped. If the second head occurs on flip number n, what is the conditional probability that the first head was on flip number $i, i = 1, \ldots, n-1$? Verify your guess analytically.

3. The joint probability mass function of X and Y, $p(x, y)$, is given by

$$
\begin{aligned}
&p(1,1) = \tfrac{1}{9}, \quad p(2,1) = \tfrac{1}{3}, \quad p(3,1) = \tfrac{1}{9}, \\
&p(1,2) = \tfrac{1}{9}, \quad p(2,2) = 0, \quad p(3,2) = \tfrac{1}{18}, \\
&p(1,3) = 0, \quad p(2,3) = \tfrac{1}{6}, \quad p(3,3) = \tfrac{1}{9}
\end{aligned}
$$

Compute $E[X|Y = i]$ for $i = 1, 2, 3$.

4. In Exercise 3, are the random variables X and Y independent?

5. An urn contains three white, six red, and five black balls. Six of these balls are randomly selected from the urn. Let X and Y denote respectively the number of white and black balls selected. Compute the conditional probability mass function of X given that $Y = 3$. Also compute $E[X|Y = 1]$.

***6.** Repeat Exercise 5 but under the assumption that when a ball is selected its color is noted, and it is then replaced in the urn before the next selection is made.

7. Suppose $p(x, y, z)$, the joint probability mass function of the random variables X, Y, and Z, is given by

$$p(1,1,1) = \tfrac{1}{8}, \quad p(2,1,1) = \tfrac{1}{4},$$
$$p(1,1,2) = \tfrac{1}{8}, \quad p(2,1,2) = \tfrac{3}{16},$$
$$p(1,2,1) = \tfrac{1}{16}, \quad p(2,2,1) = 0,$$
$$p(1,2,2) = 0, \quad p(2,2,2) = \tfrac{1}{4}$$

What is $E[X|Y=2]$? What is $E[X|Y=2, Z=1]$?

8. An unbiased die is successively rolled. Let X and Y denote, respectively, the number of rolls necessary to obtain a six and a five. Find (a) $E[X]$, (b) $E[X|Y=1]$, (c) $E[X|Y=5]$.

9. If Z_1 and Z_2 are independent standard normal random variables, find the conditional density function of Z_1 given that $Z_1 + Z_2 = x$.

10. Let $X_1, \ldots, X_n$ be independent uniform $(0, 1)$ random variables. Find the conditional density of X_1 given that it is not the smallest of the n values.

11. The joint density of X and Y is

$$f(x,y) = \frac{(y^2 - x^2)}{8} e^{-y}, \quad 0 < y < \infty, \quad -y \leqslant x \leqslant y$$

Show that $E[X|Y=y] = 0$.

12. The joint density of X and Y is given by

$$f(x,y) = \frac{e^{-x/y} e^{-y}}{y}, \quad 0 < x < \infty, \quad 0 < y < \infty$$

Show $E[X|Y=y] = y$.

***13.** Let X be exponential with mean $1/\lambda$; that is,

$$f_X(x) = \lambda e^{-\lambda x}, \quad 0 < x < \infty$$

Find $E[X|X > 1]$.

14. Let X be uniform over $(0, 1)$. Find $E[X|X < \frac{1}{2}]$.

15. The joint density of X and Y is given by

$$f(x,y) = \frac{e^{-y}}{y}, \quad 0 < x < y, \quad 0 < y < \infty$$

Compute $E[X^2|Y=y]$.

16. The random variables X and Y are said to have a bivariate normal distribution if their joint density function is given by

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)} \times \left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 - \frac{2\rho(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} + \left(\frac{y-\mu_y}{\sigma_y}\right)^2\right]\right\}$$

for $-\infty < x < \infty, -\infty < y < \infty$, where $\sigma_x, \sigma_y, \mu_x, \mu_y$, and ρ are constants such that $-1 < \rho < 1, \sigma_x > 0, \sigma_y > 0, -\infty < \mu_x < \infty, -\infty < \mu_y < \infty$.

(a) Show that X is normally distributed with mean μ_x and variance σ_x^2, and Y is normally distributed with mean μ_y and variance σ_y^2.

(b) Show that the conditional density of X given that $Y = y$ is normal with mean $\mu_x + (\rho\sigma_x/\sigma_y)(y - \mu_y)$ and variance $\sigma_x^2(1-\rho^2)$.

The quantity ρ is called the correlation between X and Y. It can be shown that

$$\rho = \frac{E[(X-\mu_x)(Y-\mu_y)]}{\sigma_x\sigma_y} = \frac{\text{Cov}(X, Y)}{\sigma_x\sigma_y}$$

17. Let Y be a gamma random variable with parameters (s, α). That is, its density is

$$f_Y(y) = Ce^{-\alpha y}y^{s-1}, \quad y > 0$$

where C is a constant that does not depend on y. Suppose also that the conditional distribution of X given that $Y = y$ is Poisson with mean y. That is,

$$P\{X = i | Y = y\} = e^{-y}y^i/i!, \quad i \geqslant 0$$

Show that the conditional distribution of Y given that $X = i$ is the gamma distribution with parameters $(s + i, \alpha + 1)$.

18. Let $X_1, \ldots, X_n$ be independent random variables having a common distribution function that is specified up to an unknown parameter θ. Let $T = T(\mathbf{X})$ be a function of the data $\mathbf{X} = (X_1, \ldots, X_n)$. If the conditional distribution of $X_1, \ldots, X_n$ given $T(\mathbf{X})$ does not depend on θ then $T(\mathbf{X})$ is said to be a *sufficient statistic* for θ. In the following cases, show that $T(\mathbf{X}) = \sum_{i=1}^n X_i$ is a sufficient statistic for θ.

(a) The X_i are normal with mean θ and variance 1.

(b) The density of X_i is $f(x) = \theta e^{-\theta x}, x > 0$.

(c) The mass function of X_i is $p(x) = \theta^x(1-\theta)^{1-x}, x = 0, 1, 0 < \theta < 1$.

(d) The X_i are Poisson random variables with mean θ.

***19.** Prove that if X and Y are jointly continuous, then

$$E[X] = \int_{-\infty}^{\infty} E[X|Y = y]f_Y(y)\,dy$$

20. There are 3 coins which when flipped come up heads, respectively, with probabilities 1/4, 1/2, 3/4. One of these coins is randomly chosen and continually flipped.

(a) Find the expected number of flips until the first head.
(b) Find the mean number of heads in the first 8 flips.

21. Consider Example 3.12, which refers to a miner trapped in a mine. Let N denote the total number of doors selected before the miner reaches safety. Also, let T_i denote the travel time corresponding to the ith choice, $i \geqslant 1$. Again let X denote the time when the miner reaches safety.
(a) Give an identity that relates X to N and the T_i.
(b) What is $E[N]$?
(c) What is $E[T_N]$?
(d) What is $E[\sum_{i=1}^{N} T_i | N = n]$?
(e) Using the preceding, what is $E[X]$?

22. Suppose that independent trials, each of which is equally likely to have any of m possible outcomes, are performed until the same outcome occurs k consecutive times. If N denotes the number of trials, show that

$$E[N] = \frac{m^k - 1}{m - 1}$$

Some people believe that the successive digits in the expansion of $\pi = 3.14159\ldots$ are "uniformly" distributed. That is, they believe that these digits have all the appearance of being independent choices from a distribution that is equally likely to be any of the digits from 0 through 9. Possible evidence against this hypothesis is the fact that starting with the 24,658,601st digit there is a run of nine successive 7s. Is this information consistent with the hypothesis of a uniform distribution?
To answer this, we note from the preceding that if the uniform hypothesis were correct, then the expected number of digits until a run of nine of the same value occurs is

$$(10^9 - 1)/9 = 111{,}111{,}111$$

Thus, the actual value of approximately 25 million is roughly 22 percent of the theoretical mean. However, it can be shown that under the uniformity assumption the standard deviation of N will be approximately equal to the mean. As a result, the observed value is approximately 0.78 standard deviations less than its theoretical mean and is thus quite consistent with the uniformity assumption.

***23.** A coin having probability p of coming up heads is successively flipped until two of the most recent three flips are heads. Let N denote the number of flips. (Note that if the first two flips are heads, then $N = 2$.) Find $E[N]$.

24. A coin, having probability p of landing heads, is continually flipped until at least one head and one tail have been flipped.
(a) Find the expected number of flips needed.
(b) Find the expected number of flips that land on heads.
(c) Find the expected number of flips that land on tails.

(d) Repeat part (a) in the case where flipping is continued until a total of at least two heads and one tail have been flipped.

25. Independent trials, resulting in one of the outcomes 1, 2, 3 with respective probabilities p_1, p_2, p_3, $\sum_{i=1}^{3} p_i = 1$, are performed.

(a) Let N denote the number of trials needed until the initial outcome has occurred exactly 3 times. For instance, if the trial results are 3, 2, 1, 2, 3, 2, 3 then $N = 7$. Find $E[N]$.

(b) Find the expected number of trials needed until both outcome 1 and outcome 2 have occurred.

26. You have two opponents with whom you alternate play. Whenever you play A, you win with probability p_A; whenever you play B, you win with probability p_B, where $p_B > p_A$. If your objective is to minimize the expected number of games you need to play to win two in a row, should you start with A or with B?

Hint: Let $E[N_i]$ denote the mean number of games needed if you initially play i. Derive an expression for $E[N_A]$ that involves $E[N_B]$; write down the equivalent expression for $E[N_B]$ and then subtract.

27. A coin that comes up heads with probability p is continually flipped until the pattern T, T, H appears. (That is, you stop flipping when the most recent flip lands heads, and the two immediately preceding it lands tails.) Let X denote the number of flips made, and find $E[X]$.

28. Polya's urn model supposes that an urn initially contains r red and b blue balls. At each stage a ball is randomly selected from the urn and is then returned along with m other balls of the same color. Let X_k be the number of red balls drawn in the first k selections.

(a) Find $E[X_1]$.

(b) Find $E[X_2]$.

(c) Find $E[X_3]$.

(d) Conjecture the value of $E[X_k]$, and then verify your conjecture by a conditioning argument.

(e) Give an intuitive proof for your conjecture.

Hint: Number the initial r red and b blue balls, so the urn contains one type i red ball, for each $i = 1, \ldots, r$; as well as one type j blue ball, for each $j = 1, \ldots, b$. Now suppose that whenever a red ball is chosen it is returned along with m others of the same type, and similarly whenever a blue ball is chosen it is returned along with m others of the same type. Now, use a symmetry argument to determine the probability that any given selection is red.

29. Two players take turns shooting at a target, with each shot by player i hitting the target with probability $p_i, i = 1, 2$. Shooting ends when two consecutive shots hit the target. Let μ_i denote the mean number of shots taken when player i shoots first, $i = 1, 2$.

(a) Find μ_1 and μ_2.

(b) Let h_i denote the mean number of times that the target is hit when player i shoots first, $i = 1, 2$. Find h_1 and h_2.

30. Let $X_i, i \geqslant 0$ be independent and identically distributed random variables with probability mass function

$$p(j) = P\{X_i = j\}, \quad j = 1, \ldots, m, \quad \sum_{j=1}^{m} P(j) = 1$$

Find $E[N]$, where $N = \min\{n > 0 : X_n = X_0\}$.

31. Each element in a sequence of binary data is either 1 with probability p or 0 with probability $1 - p$. A maximal subsequence of consecutive values having identical outcomes is called a run. For instance, if the outcome sequence is $1, 1, 0, 1, 1, 1, 0$, the first run is of length 2, the second is of length 1, and the third is of length 3.

(a) Find the expected length of the first run.

(b) Find the expected length of the second run.

32. Independent trials, each resulting in success with probability p, are performed.

(a) Find the expected number of trials needed for there to have been both at least n successes and at least m failures.

Hint: Is it useful to know the result of the first $n + m$ trials?

(b) Find the expected number of trials needed for there to have been either at least n successes or at least m failures.

Hint: Make use of the result from part (a).

33. If R_i denotes the random amount that is earned in period i, then $\sum_{i=1}^{\infty} \beta^{i-1} R_i$, where $0 < \beta < 1$ is a specified constant, is called the total discounted reward with discount factor β. Let T be a geometric random variable with parameter $1 - \beta$ that is independent of the R_i. Show that the expected total discounted reward is equal to the expected total (undiscounted) reward earned by time T. That is, show that

$$E\left[\sum_{i=1}^{\infty} \beta^{i-1} R_i\right] = E\left[\sum_{i=1}^{T} R_i\right]$$

34. A set of n dice is thrown. All those that land on six are put aside, and the others are again thrown. This is repeated until all the dice have landed on six. Let N denote the number of throws needed. (For instance, suppose that $n = 3$ and that on the initial throw exactly two of the dice land on six. Then the other die will be thrown, and if it lands on six, then $N = 2$.) Let $m_n = E[N]$.

(a) Derive a recursive formula for m_n and use it to calculate $m_i, i = 2, 3, 4$ and to show that $m_5 \approx 13.024$.

(b) Let X_i denote the number of dice rolled on the ith throw. Find $E[\sum_{i=1}^{N} X_i]$.

35. Consider n multinomial trials, where each trial independently results in outcome i with probability p_i, $\sum_{i=1}^{k} p_i = 1$. With X_i equal to the number of trials that result in outcome i, find $E[X_1 | X_2 > 0]$.

36. Let $p_0 = P\{X = 0\}$ and suppose that $0 < p_0 < 1$. Let $\mu = E[X]$ and $\sigma^2 = \text{Var}(X)$.
- **(a)** Find $E[X|X \neq 0]$.
- **(b)** Find $\text{Var}(X|X \neq 0)$.

37. A manuscript is sent to a typing firm consisting of typists A, B, and C. If it is typed by A, then the number of errors made is a Poisson random variable with mean 2.6; if typed by B, then the number of errors is a Poisson random variable with mean 3; and if typed by C, then it is a Poisson random variable with mean 3.4. Let X denote the number of errors in the typed manuscript. Assume that each typist is equally likely to do the work.
- **(a)** Find $E[X]$.
- **(b)** Find $\text{Var}(X)$.

38. Suppose Y is uniformly distributed on $(0, 1)$, and that the conditional distribution of X given that $Y = y$ is uniform on $(0, y)$. Find $E[X]$ and $\text{Var}(X)$.

39. A deck of n cards, numbered 1 through n, is randomly shuffled so that all $n!$ possible permutations are equally likely. The cards are then turned over one at a time until card number 1 appears. These upturned cards constitute the first cycle. We now determine (by looking at the upturned cards) the lowest numbered card that has not yet appeared, and we continue to turn the cards face up until that card appears. This new set of cards represents the second cycle. We again determine the lowest numbered of the remaining cards and turn the cards until it appears, and so on until all cards have been turned over. Let m_n denote the mean number of cycles.
- **(a)** Derive a recursive formula for m_n in terms of $m_k, k = 1, \ldots, n-1$.
- **(b)** Starting with $m_0 = 0$, use the recursion to find m_1, m_2, m_3, and m_4.
- **(c)** Conjecture a general formula for m_n.
- **(d)** Prove your formula by induction on n. That is, show it is valid for $n = 1$, then assume it is true for any of the values $1, \ldots, n-1$ and show that this implies it is true for n.
- **(e)** Let X_i equal 1 if one of the cycles ends with card i, and let it equal 0 otherwise, $i = 1, \ldots, n$. Express the number of cycles in terms of these X_i.
- **(f)** Use the representation in part (e) to determine m_n.
- **(g)** Are the random variables $X_1, \ldots, X_n$ independent? Explain.
- **(h)** Find the variance of the number of cycles.

40. A prisoner is trapped in a cell containing three doors. The first door leads to a tunnel that returns him to his cell after two days of travel. The second leads to a tunnel that returns him to his cell after three days of travel. The third door leads immediately to freedom.
- **(a)** Assuming that the prisoner will always select doors 1, 2, and 3 with probabilities 0.5, 0.3, 0.2, what is the expected number of days until he reaches freedom?
- **(b)** Assuming that the prisoner is always equally likely to choose among those doors that he has not used, what is the expected number of days until he reaches freedom? (In this version, for instance, if the prisoner

initially tries door 1, then when he returns to the cell, he will now select only from doors 2 and 3.)

(c) For parts (a) and (b) find the variance of the number of days until the prisoner reaches freedom.

41. Workers $1, \ldots, n$ are currently idle. Suppose that each worker, independently, has probability p of being eligible for a job, and that a job is equally likely to be assigned to any of the workers that are eligible for it (if none are eligible, the job is rejected). Find the probability that the next job is assigned to worker 1.

***42.** If $X_i, i = 1, \ldots, n$ are independent normal random variables, with X_i having mean μ_i and variance 1, then the random variable $\sum_{i=1}^{n} X_i^2$ is said to be a *noncentral chi-squared* random variable.

(a) if X is a normal random variable having mean μ and variance 1 show, for $|t| < 1/2$, that the moment generating function of X^2 is

$$(1-2t)^{-1/2} e^{\frac{t\mu^2}{1-2t}}$$

(b) Derive the moment generating function of the noncentral chi-squared random variable $\sum_{i=1}^{n} X_i^2$, and show that its distribution depends on the sequence of means $\mu_1, \ldots, \mu_n$ only through the sum of their squares. As a result, we say that $\sum_{i=1}^{n} X_i^2$ is a noncentral chi-squared random variable with parameters n and $\theta = \sum_{i=1}^{n} \mu_i^2$.

(c) If all $\mu_i = 0$, then $\sum_{i=1}^{n} X_i^2$ is called a chi-squared random variable with n degrees of freedom. Determine, by differentiating its moment generating function, its expected value and variance.

(d) Let K be a Poisson random variable with mean $\theta/2$, and suppose that conditional on $K = k$, the random variable W has a chi-squared distribution with $n + 2k$ degrees of freedom. Show, by computing its moment generating function, that W is a noncentral chi-squared random variable with parameters n and θ.

(e) Find the expected value and variance of a noncentral chi-squared random variable with parameters n and θ.

***43.** For $P(Y \in A) > 0$, show that

$$E[X|Y \in A] = \frac{E[XI\{Y \in A\}]}{P(Y \in A)}$$

where $I\{B\}$ is the indicator variable of the event B, equal to 1 if B occurs and to 0 otherwise.

44. The number of customers entering a store on a given day is Poisson distributed with mean $\lambda = 10$. The amount of money spent by a customer is uniformly distributed over (0, 100). Find the mean and variance of the amount of money that the store takes in on a given day.

45. An individual traveling on the real line is trying to reach the origin. However, the larger the desired step, the greater is the variance in the result of that step.

Specifically, whenever the person is at location x, he next moves to a location having mean 0 and variance βx^2. Let X_n denote the position of the individual after having taken n steps. Supposing that $X_0 = x_0$, find

(a) $E[X_n]$;

(b) $\text{Var}(X_n)$.

46. **(a)** Show that

$$\text{Cov}(X, Y) = \text{Cov}(X, E[Y|X])$$

(b) Suppose, that, for constants a and b,

$$E[Y|X] = a + bX$$

Show that

$$b = \text{Cov}(X, Y)/\text{Var}(X)$$

***47.** If $E[Y|X] = 1$, show that

$$\text{Var}(XY) \geqslant \text{Var}(X)$$

48. Suppose that we want to predict the value of a random variable X by using one of the predictors $Y_1, \ldots, Y_n$, each of which satisfies $E[Y_i|X] = X$. Show that the predictor Y_i that minimizes $E[(Y_i - X)^2]$ is the one whose variance is smallest.

Hint: Compute $\text{Var}(Y_i)$ by using the conditional variance formula.

49. A and B play a series of games with A winning each game with probability p. The overall winner is the first player to have won two more games than the other.

(a) Find the probability that A is the overall winner.

(b) Find the expected number of games played.

50. Suppose that N, the number of flips made of a coin that comes up heads with probability p, is a geometric random variable with parameter α, independent of the results of the flips. Let A be the event that all flips land heads.

(a) Find $P(A)$ by conditioning on N.

(b) Find $P(A)$ by conditioning on the result of the first flip.

51. If X is geometric with parameter p, find the probability that X is even.

52. Each applicant has a score. If there are a total of n applicants then each applicant whose score is above s_n is accepted, where $s_1 = .2$, $s_2 = .4$, $s_n = .5$, $n \geq 3$. Suppose the scores of the applicants are independent uniform $(0, 1)$ random variables and are independent of N, the number of applicants, which is Poisson distributed with mean 2. Let X denote the number of applicants that are accepted. Derive expressions for

(a) $P(X = 0)$.

(b) $E[X]$.

***53.** Suppose X is a Poisson random variable with mean λ. The parameter λ is itself a random variable whose distribution is exponential with mean 1. Show that $P\{X = n\} = (\frac{1}{2})^{n+1}$.

54. Independent trials, each resulting in a success with probability p, are performed until k consecutive successful trials have occurred. Let X be the total number of successes in these trial, and let $P_n = P(X = n)$.

(a) Find P_k.

(b) Derive a recursive equation for the $P_n, n \geq k$, by imagining that the trials continue forever and conditioning on the time of the first failure.

(c) Verify your answer in part (a) by solving the recursion for P_k.

(d) When $p = .6, k = 3$, find P_8.

55. In the preceding problem let $M_k = E[X]$. Derive a recursive equation for M_k and then solve.

Hint: Start by writing $X_k = X_{k-1} + A_{k-1,k}$, where X_i is the total number of successes attained up to the first time there have been i consecutive successes, and $A_{k-1,k}$ is the additional number of successes after there have been $k - 1$ successes in a row until there have been k successes in a row.

56. Data indicate that the number of traffic accidents in Berkeley on a rainy day is a Poisson random variable with mean 9, whereas on a dry day it is a Poisson random variable with mean 3. Let X denote the number of traffic accidents tomorrow. If it will rain tomorrow with probability 0.6, find

(a) $E[X]$;

(b) $P\{X = 0\}$;

(c) $\text{Var}(X)$.

57. The number of storms in the upcoming rainy season is Poisson distributed but with a parameter value that is uniformly distributed over $(0, 5)$. That is, Λ is uniformly distributed over $(0, 5)$, and given that $\Lambda = \lambda$, the number of storms is Poisson with mean λ. Find the probability there are at least three storms this season.

***58.** Suppose that the conditional distribution of N, given that $Y = y$, is Poisson with mean y. Further suppose that Y is a gamma random variable with parameters (r, λ), where r is a positive integer. That is, suppose that

$$P(N = n|Y = y) = e^{-y}\frac{y^n}{n!}$$

and

$$f_Y(y) = \frac{\lambda e^{-\lambda y}(\lambda y)^{r-1}}{(r-1)!}, \quad y > 0$$

(a) Find $E[N]$.

(b) Find $\text{Var}(N)$.

(c) Find $P(N = n)$

(d) Using (c), argue that N is distributed as the total number of failures before the rth success when each trial is independently a success with probability $p = \frac{\lambda}{1+\lambda}$.

59. Suppose each new coupon collected is, independent of the past, a type i coupon with probability p_i. A total of n coupons is to be collected. Let A_i be the event that there is at least one type i in this set. For $i \neq j$, compute $P(A_i A_j)$ by

(a) conditioning on N_i, the number of type i coupons in the set of n coupons;

(b) conditioning on F_i, the first time a type i coupon is collected;

(c) using the identity $P(A_i \cup A_j) = P(A_i) + P(A_j) - P(A_i A_j)$.

***60.** Two players alternate flipping a coin that comes up heads with probability p. The first one to obtain a head is declared the winner. We are interested in the probability that the first player to flip is the winner. Before determining this probability, which we will call $f(p)$, answer the following questions.

(a) Do you think that $f(p)$ is a monotone function of p? If so, is it increasing or decreasing?

(b) What do you think is the value of $\lim_{p\to 1} f(p)$?

(c) What do you think is the value of $\lim_{p\to 0} f(p)$?

(d) Find $f(p)$.

61. Suppose in Exercise 29 that the shooting ends when the target has been hit twice. Let m_i denote the mean number of shots needed for the first hit when player i shoots first, $i = 1, 2$. Also, let $P_i, i = 1, 2$, denote the probability that the first hit is by player 1, when player i shoots first.

(a) Find m_1 and m_2.

(b) Find P_1 and P_2.

For the remainder of the problem, assume that player 1 shoots first.

(c) Find the probability that the final hit was by 1.

(d) Find the probability that both hits were by 1.

(e) Find the probability that both hits were by 2.

(f) Find the mean number of shots taken.

62. A, B, and C are evenly matched tennis players. Initially A and B play a set, and the winner then plays C. This continues, with the winner always playing the waiting player, until one of the players has won two sets in a row. That player is then declared the overall winner. Find the probability that A is the overall winner.

63. Suppose there are n types of coupons, and that the type of each new coupon obtained is independent of past selections and is equally likely to be any of the n types. Suppose one continues collecting until a complete set of at least one of each type is obtained.

(a) Find the probability that there is exactly one type i coupon in the final collection.

Hint: Condition on T, the number of types that are collected before the first type i appears.

(b) Find the expected number of types that appear exactly once in the final collection.

64. A and B roll a pair of dice in turn, with A rolling first. A's objective is to obtain a sum of 6, and B's is to obtain a sum of 7. The game ends when either player reaches his or her objective, and that player is declared the winner.
- **(a)** Find the probability that A is the winner.
- **(b)** Find the expected number of rolls of the dice.
- **(c)** Find the variance of the number of rolls of the dice.

65. The number of red balls in an urn that contains n balls is a random variable that is equally likely to be any of the values $0, 1, \ldots, n$. That is,

$$P\{i \text{ red}, n-i \text{ non-red}\} = \frac{1}{n+1}, \quad i = 0, \ldots, n$$

The n balls are then randomly removed one at a time. Let Y_k denote the number of red balls in the first k selections, $k = 1, \ldots, n$.
- **(a)** Find $P\{Y_n = j\}$, $j = 0, \ldots, n$.
- **(b)** Find $P\{Y_{n-1} = j\}$, $j = 0, \ldots, n$.
- **(c)** What do you think is the value of $P\{Y_k = j\}$, $j = 0, \ldots, n$?
- **(d)** Verify your answer to part (c) by a backwards induction argument. That is, check that your answer is correct when $k = n$, and then show that whenever it is true for k it is also true for $k-1$, $k = 1, \ldots, n$.

66. The number of goals that J scores in soccer games that her team wins is Poisson with mean 2, while the number she scores in games that her team loses is Poisson with mean 1. Assume that J's team wins each game they play with probability p.
- **(a)** Find the expected number of goals that J scores in her next 3 games.
- **(b)** Find the probability that she scores a total of n goals in her next 3 games.

***67.** A coin having probability p of coming up heads is continually flipped. Let $P_j(n)$ denote the probability that a run of j successive heads occurs within the first n flips.
- **(a)** Argue that

$$P_j(n) = P_j(n-1) + p^j(1-p)[1 - P_j(n-j-1)]$$

- **(b)** By conditioning on the first non-head to appear, derive another equation relating $P_j(n)$ to the quantities $P_j(n-k)$, $k = 1, \ldots, j$.

68. If the level of infection of a tree is x, $0 \le x \le 1$, then each cure attempt will independently be successful with probability $1 - x$. Consider a tree whose infection level, call it L, is assumed to be the value of a uniform (0, 1) random variable.
- **(a)** What is the probability that a single attempt will result in a cure.
- **(b)** Find the probability that the first two cure attempts are unsuccessful.
- **(c)** Find the conditional expected value of L given that it took 3 attempts to cure the tree.

69. In the match problem, say that (i, j), $i < j$, is a pair if i chooses j's hat and j chooses i's hat.
- **(a)** Find the expected number of pairs.

(b) Let Q_n denote the probability that there are no pairs, and derive a recursive formula for Q_n in terms of Q_j, $j < n$.

Hint: Use the cycle concept.

(c) Use the recursion of part (b) to find Q_8.

70. Let N denote the number of cycles that result in the match problem.

(a) Let $M_n = E[N]$, and derive an equation for M_n in terms of $M_1, \ldots, M_{n-1}$.

(b) Let C_j denote the size of the cycle that contains person j. Argue that

$$N = \sum_{j=1}^{n} 1/C_j$$

and use the preceding to determine $E[N]$.

(c) Find the probability that persons $1, 2, \ldots, k$ are all in the same cycle.

(d) Find the probability that $1, 2, \ldots, k$ is a cycle.

71. Use Eq. (3.13) to obtain Eq. (3.9).

Hint: First multiply both sides of Eq. (3.13) by n, then write a new equation by replacing n with $n - 1$, and then subtract the former from the latter.

72. In Example 3.29 show that the conditional distribution of N given that $U_1 = y$ is the same as the conditional distribution of M given that $U_1 = 1 - y$. Also, show that

$$E[N|U_1 = y] = E[M|U_1 = 1 - y] = 1 + e^y$$

***73.** Suppose that we continually roll a die until the sum of all throws exceeds 100. What is the most likely value of this total when you stop?

74. There are five components. The components act independently, with component i working with probability p_i, $i = 1, 2, 3, 4, 5$. These components form a system as shown in Fig. 3.7.

The system is said to work if a signal originating at the left end of the diagram can reach the right end, where it can pass through a component only if that component is working. (For instance, if components 1 and 4 both work, then the system also works.) What is the probability that the system works?

75. This problem will present another proof of the ballot problem of Example 3.28.

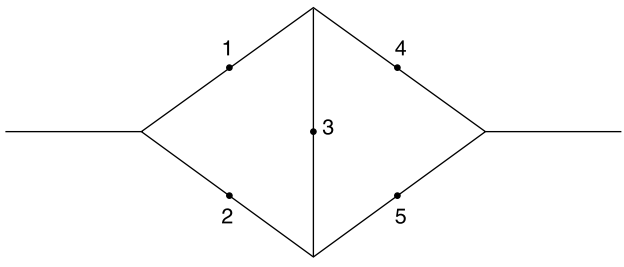


Figure 3.7

(a) Argue that

$$P_{n,m} = 1 - P\{A \text{ and } B \text{ are tied at some point}\}$$

(b) Explain why

$$P\{A \text{ receives first vote and they are eventually tied}\}$$
$$= P\{B \text{ receives first vote and they are eventually tied}\}$$

Hint: Any outcome in which they are eventually tied with A receiving the first vote corresponds to an outcome in which they are eventually tied with B receiving the first vote. Explain this correspondence.

(c) Argue that $P\{\text{eventually tied}\} = 2m/(n+m)$, and conclude that $P_{n,m} = (n-m)/(n+m)$.

76. Consider a gambler who on each bet either wins 1 with probability 18/38 or loses 1 with probability 20/38. (These are the probabilities if the bet is that a roulette wheel will land on a specified color.) The gambler will quit either when he or she is winning a total of 5 or after 100 plays. What is the probability he or she plays exactly 15 times?

77. Show that

(a) $E[XY|Y = y] = yE[X|Y = y]$

(b) $E[g(X, Y)|Y = y] = E[g(X, y)|Y = y]$

(c) $E[XY] = E[YE[X|Y]]$

78. In the ballot problem (Example 3.28), compute $P\{A \text{ is never behind}\}$.

79. An urn contains n white and m black balls that are removed one at a time. If $n > m$, show that the probability that there are always more white than black balls in the urn (until, of course, the urn is empty) equals $(n-m)/(n+m)$. Explain why this probability is equal to the probability that the set of withdrawn balls always contains more white than black balls. (This latter probability is $(n-m)/(n+m)$ by the ballot problem.)

80. A coin that comes up heads with probability p is flipped n consecutive times. What is the probability that starting with the first flip there are always more heads than tails that have appeared?

81. Let $X_i, i \geqslant 1$, be independent uniform (0, 1) random variables, and define N by

$$N = \min\{n: X_n < X_{n-1}\}$$

where $X_0 = x$. Let $f(x) = E[N]$.

(a) Derive an integral equation for $f(x)$ by conditioning on X_1.

(b) Differentiate both sides of the equation derived in part (a).

(c) Solve the resulting equation obtained in part (b).

(d) For a second approach to determining $f(x)$ argue that

$$P\{N \geqslant k\} = \frac{(1-x)^{k-1}}{(k-1)!}$$

(e) Use part (d) to obtain $f(x)$.

82. Let $X_1, X_2, \ldots$ be independent continuous random variables with a common distribution function F and density $f = F'$, and for $k \geqslant 1$ let

$$N_k = \min\{n \geqslant k: X_n = k\text{th largest of } X_1, \ldots, X_n\}$$

(a) Show that $P\{N_k = n\} = \frac{k-1}{n(n-1)}$, $n \geqslant k$.

(b) Argue that

$$f_{X_{N_k}}(x) = f(x)(\bar{F}(x))^{k-1} \sum_{i=0}^{\infty} \binom{i+k-2}{i} (F(x))^i$$

(c) Prove the following identity:

$$a^{1-k} = \sum_{i=0}^{\infty} \binom{i+k-2}{i} (1-a)^i, \quad 0 < a < 1, k \geqslant 2$$

Hint: Use induction. First prove it when $k = 2$, and then assume it for k. To prove it for $k+1$, use the fact that

$$\sum_{i=1}^{\infty} \binom{i+k-1}{i} (1-a)^i = \sum_{i=1}^{\infty} \binom{i+k-2}{i} (1-a)^i + \sum_{i=1}^{\infty} \binom{i+k-2}{i-1} (1-a)^i$$

where the preceding used the combinatorial identity

$$\binom{m}{i} = \binom{m-1}{i} + \binom{m-i}{i-1}$$

Now, use the induction hypothesis to evaluate the first term on the right side of the preceding equation.

(d) Conclude that X_{N_k} has distribution F.

83. An urn contains n balls, with ball i having weight $w_i, i = 1, \ldots, n$. The balls are withdrawn from the urn one at a time according to the following scheme: When S is the set of balls that remains, ball $i, i \in S$, is the next ball withdrawn with probability $w_i / \sum_{j \in S} w_j$. Find the expected number of balls that are withdrawn before ball $i, i = 1, \ldots, n$.

84. Suppose in Example 3.33 that a point is only won if the winner of the rally was the server of that rally.

(a) If A is currently serving, what is the probability that A wins the next point?

(b) Explain how to obtain the final score probabilities.

85. In the list problem, when the P_i are known, show that the best ordering (best in the sense of minimizing the expected position of the element requested) is to place the elements in decreasing order of their probabilities. That is, if $P_1 > P_2 > \cdots > P_n$, show that $1, 2, \ldots, n$ is the best ordering.

86. Consider the random graph of Section 3.6.2 when $n = 5$. Compute the probability distribution of the number of components and verify your solution by using it to compute $E[C]$ and then comparing your solution with

$$E[C] = \sum_{k=1}^{5} \binom{5}{k} \frac{(k-1)!}{5^k}$$

87. **(a)** From the results of Section 3.6.3 we can conclude that there are $\binom{n+m-1}{m-1}$ nonnegative integer valued solutions of the equation $x_1 + \cdots + x_m = n$. Prove this directly.

(b) How many positive integer valued solutions of $x_1 + \cdots + x_m = n$ are there?

Hint: Let $y_i = x_i - 1$.

(c) For the Bose–Einstein distribution, compute the probability that exactly k of the X_i are equal to 0.

88. In Section 3.6.3, we saw that if U is a random variable that is uniform on (0, 1) and if, conditional on $U = p$, X is binomial with parameters n and p, then

$$P\{X = i\} = \frac{1}{n+1}, \quad i = 0, 1, \ldots, n$$

For another way of showing this result, let $U, X_1, X_2, \ldots, X_n$ be independent uniform (0, 1) random variables. Define X by

$$X = \#i\colon X_i < U$$

That is, if the $n+1$ variables are ordered from smallest to largest, then U would be in position $X + 1$.

(a) What is $P\{X = i\}$?

(b) Explain how this proves the result of Section 3.6.3.

89. Let $I_1, \ldots, I_n$ be independent random variables, each of which is equally likely to be either 0 or 1. A well-known nonparametric statistical test (called the signed rank test) is concerned with determining $P_n(k)$ defined by

$$P_n(k) = P\left\{\sum_{j=1}^{n} j I_j \leqslant k\right\}$$

Justify the following formula:

$$P_n(k) = \tfrac{1}{2} P_{n-1}(k) + \tfrac{1}{2} P_{n-1}(k-n)$$

90. The number of accidents in each period is a Poisson random variable with mean 5. With $X_n, n \geq 1$, equal to the number of accidents in period n, find $E[N]$ when
(a) $N = \min(n: X_{n-2} = 2, X_{n-1} = 1, X_n = 0)$;
(b) $N = \min(n: X_{n-3} = 2, X_{n-2} = 1, X_{n-1} = 0, X_n = 2)$.

91. Find the expected number of flips of a coin, which comes up heads with probability p, that are necessary to obtain the pattern h, t, h, h, t, h, t, h.

92. The number of coins that Josh spots when walking to work is a Poisson random variable with mean 6. Each coin is equally likely to be a penny, a nickel, a dime, or a quarter. Josh ignores the pennies but picks up the other coins.
(a) Find the expected amount of money that Josh picks up on his way to work.
(b) Find the variance of the amount of money that Josh picks up on his way to work.
(c) Find the probability that Josh picks up exactly 25 cents on his way to work.

***93.** Consider a sequence of independent trials, each of which is equally likely to result in any of the outcomes $0, 1, \ldots, m$. Say that a round begins with the first trial, and that a new round begins each time outcome 0 occurs. Let N denote the number of trials that it takes until all of the outcomes $1, \ldots, m-1$ have occurred in the same round. Also, let T_j denote the number of trials that it takes until j distinct outcomes have occurred, and let I_j denote the jth distinct outcome to occur. (Therefore, outcome I_j first occurs at trial T_j.)
(a) Argue that the random vectors $(I_1, \ldots, I_m)$ and $(T_1, \ldots, T_m)$ are independent.
(b) Define X by letting $X = j$ if outcome 0 is the jth distinct outcome to occur. (Thus, $I_X = 0$.) Derive an equation for $E[N]$ in terms of $E[T_j], j = 1, \ldots, m-1$ by conditioning on X.
(c) Determine $E[T_j], j = 1, \ldots, m-1$.

Hint: See Exercise 42 of Chapter 2.
(d) Find $E[N]$.

94. Let N be a hypergeometric random variable having the distribution of the number of white balls in a random sample of size r from a set of w white and b blue balls. That is,

$$P\{N = n\} = \frac{\binom{w}{n}\binom{b}{r-n}}{\binom{w+b}{r}}$$

where we use the convention that $\binom{m}{j} = 0$ if either $j < 0$ or $j > m$. Now, consider a compound random variable $S_N = \sum_{i=1}^{N} X_i$, where the X_i are positive integer valued random variables with $\alpha_j = P\{X_i = j\}$.
(a) With M as defined as in Section 3.7, find the distribution of $M - 1$.

(b) Suppressing its dependence on b, let $P_{w,r}(k) = P\{S_N = k\}$, and derive a recursion equation for $P_{w,r}(k)$.

(c) Use the recursion of (b) to find $P_{w,r}(2)$.

95. For the left skip free random walk of Section 3.6.6 let $\beta = P(S_n \leq 0 \text{ for all } n)$ be the probability that the walk is never positive. Find β when $E[X_i] < 0$.

96. Consider a large population of families, and suppose that the number of children in the different families are independent Poisson random variables with mean λ. Show that the number of siblings of a randomly chosen child is also Poisson distributed with mean λ.

***97.** Use the conditional variance formula to find the variance of a geometric random variable.

98. For a compound random variable $S = \sum_{i=1}^{N} X_i$, find $\text{Cov}(N, S)$.

99. Let N be the number of trials until k consecutive successes have occurred, when each trial is independently a success with probability p.

(a) What is $P(N = k)$?

(b) Argue that

$$P(N = k + r) = P(N > r - 1)qp^k, \quad r > 0$$

(c) Show that

$$1 - p^k = qp^k E[N]$$

100. In the fair gambler's ruin problem of Example 3.16, let P_i denote the probability that, starting with a fortune of i, the gambler's fortune reaches n before 0. Find P_i, $0 \leq i \leq n$.

101. For the left skip free random walk of Section 3.6.6,

(a) Show, for $0 < k \leq n$, that $P(T_k = n | S_n = -k) = k/n$.

(b) Show that part (a) implies that $P(S_j < 0,\ j = 1, \ldots, n | S_n = -k) = k/n$.

(c) Explain why part (b) implies the ballot theorem.